

Antimicrobial Peptide Design and Discovery with Convolutional and Graph Neural Networks: Homology-Aware Benchmarks, External Validation on APD6, and a Discovery Screen over Uncharacterized Proteins

MEGA-Program-27, Item 9 Part 1 (lane 09a)

September 24, 2026

Abstract

Antimicrobial peptides (AMPs) are a structurally diverse class of host-defense molecules and a candidate answer to antibiotic resistance. We study AMP recognition and design with two neural architectures — a convolutional network (PepCNN) over a joint one-hot/physicochemical sequence encoding and a graph neural network (PepGNN) over a residue graph with a shared positional adjacency — trained and evaluated on 24,628 UniProt KW-0929 AMP-annotated entries (mostly unreviewed) against 24,628 length-matched reviewed non-AMP proteins, with exact-10-mer cluster-aware splits. The positive/negative source-review status is severely confounded and the scale-up scores are descriptive only. PepCNN attains AUROC 0.947 [0.936, 0.955] under 5-fold cross-validation on a curated 2,732-sequence pilot and 0.921 [0.913, 0.929] on the held-out large-scale test split; a random-forest baseline on dipeptide composition is competitive on the pilot (0.935) but the convolutional model dominates at scale and on external validation against the APD6 2024 natural AMP list, but this test does not remove the training-source confound. We prove fixed-temperature detailed balance of a simulated-annealing transition used to propose untested candidate sequences, derive the permutation and bootstrap procedures used for significance, and report an in-silico discovery screen over 852 reviewed uncharacterized proteins of 10–60 residues. On the published AMP Scanner Vr.2 benchmark our stacked ensemble is near the original model on its own test partition but below the later ACEP model. We additionally audit an updated Feb2020 production dataset with 10-fold cross-validation; comparability limits and detailed results are reported. Eight screened proteins and one annealed design are named as testable candidates with quantified pairwise 3-mer similarity checks against APD6 and a UniProt KW-0929 subsample, not established antimicrobial discoveries. All results are reproducible from the accompanying repository; claims are limited to what the data support.

1 Introduction

The spread of antibiotic resistance has renewed interest in antimicrobial peptides (AMPs), short cationic and often amphipathic sequences that disrupt microbial membranes and are harder for bacteria to evade than small-molecule antibiotics [1, 11]. Databases such as the APD (now APD6) [1] and DBAASP [2] curate thousands of experimentally characterized AMPs, and UniProt keyword KW-0929 annotates both reviewed and unreviewed antimicrobial proteins at scale [3]. Computational screening and design of AMPs is attractive because the sequence space is enormous: even at length 25 there are $20^{25} \approx 3.4 \times 10^{32}$ candidates, of which only a minuscule fraction has been assayed.

This paper, item 9 part 1 of the MEGA-Program-27 series, asks three questions. (1) How well do convolutional and graph neural networks recognize AMPs when evaluation is protected against homology leakage — the dominant confound in peptide classification benchmarks? (2) Do models trained on UniProt KW-0929 generalize to an independently curated resource, APD6? (3) Can the trained models prioritize plausible novel AMPs, both by free design (simulated annealing against the learned score) and by screening uncharacterized proteins?

1.1 Biological setting

AMPs are produced across all domains of life. The canonical family — short, net-cationic, amphipathic peptides such as magainins, LL-37, and the defensins — partitions into microbial membranes and kills by permeabilization, but the known universe also includes anionic AMPs, glycine-rich peptides, proline-rich peptides that enter cells and inhibit translation, and larger disulfide-stabilized folds. This mechanistic variety is the central recognition challenge: there is no single sequence signature, and any single-feature rule (charge, hydrophobic moment, length) fails on whole families. It is also why we treat sensitivity numbers honestly: a model trained on a corpus dominated by short cationic peptides should not be expected to score disulfide-locked anionic families highly, and ours does not (Section 7.3).

1.2 Contributions

1. An auditable, exact-10-mer cluster-split benchmark with a subsequently identified source-review confound: 24,628 UniProt KW-0929 positives (mostly unreviewed) against 24,628 length-matched reviewed negatives, clustered by shared exact 10-mers into 12,110 components and split 80/10/10 by cluster, with all fetch URLs, hashes, and timestamps recorded.
2. A paired comparison of two neural architectures and two composition baselines under this split, with bootstrap intervals and permutation tests computed by one shared harness: PepCNN attains descriptive AUROC 0.921 [0.913, 0.929] on the confounded held-out sample, ahead of the tested baselines; the graph model is a reported negative result.
3. An APD6 2024 external-set comparison after exact-10-mer filtering: PepCNN scores AUROC 0.926 [0.918, 0.934] and sensitivity 0.601 at a 5% false-positive rate, but review-status confounding in training limits its biological interpretation.

4. A simulated-annealing designer with a fixed-temperature detailed-balance proof producing ranked sequence candidates, and a discovery screen over 852 reviewed uncharacterized proteins producing 25 ranked hypotheses, all clearly labeled as in-silico.
5. A reproducible software artifact: the pepdesign package, multiple study scripts, a 27-test hermetic suite, and reproducible benchmark and candidate tables derived from checked-in result files.

1.3 Scope

This paper is item 9 part 1 of the MEGA-Program-27 series and is deliberately bounded: recognition, design, and screening of AMPs. Activity prediction against specific strains, toxicity estimation, secondary-structure validation, and wet-lab follow-up are out of scope and are the natural subjects of subsequent items.

1.4 Notation and organization

Table 1: Notation used throughout.

Symbol	Meaning
s, ℓ	a peptide sequence and its length
L	encoding cap (150; pilot 60; full Feb2020 audit 200)
$K_k(s)$	set of contiguous k -mers of s
$E(s)$	encoded sequence, $\mathbb{R}^{L \times (20+d)}$
$\tilde{A}$	symmetric-normalized shared adjacency
$\hat{p}, f(s)$	sigmoid classifier output and ensemble score
ω	positive-class weight n_-/n_+
$\pi_T(s), T$	fixed-temperature Boltzmann measure and temperature
$J(\cdot, \cdot)$	Jaccard index
B	resampling count (bootstrap or permutation)

Section 2 situates the work; Section 3 defines the data, clustering, and splits; Section 4 defines the models, objective, designer, and statistical procedures, with proofs; Section 5 fixes the evaluation metrics; Later sections report the benchmark studies, candidate screen, signal ablation, threats, reproducibility audit and software artifact. Appendices give hyperparameters, provenance and design/screen listings.

2 Related Work

The APD has been curated since 2003; its third version, APD3, organized AMP annotation and search [1], and the 2024 APD6 release provides downloadable FASTA lists of natural AMPs with known activity. DBAASP adds experimentally measured activities and targets [2]. Machine-learning AMP classifiers span support-vector machines and random forests on composition features (e.g., CAMP [4]) to recurrent and convolutional architectures such as AMPScanner [6]. Most

published benchmarks split sequences randomly, which leaks near-duplicates across train and test; we instead cluster on shared k -mers and split clusters. Our graph model is related to message-passing networks [7] with a fixed chain-adjacency structure rather than a learned or contact-derived graph.

2.1 AMP databases

The APD has tracked the natural AMP literature since 2003; the APD3 release [1] introduced a unified search and annotation interface, and the current APD6 site distributes curated FASTA lists, including the 2024 natural-AMP list (3,306 entries) used here as external validation. DBAASP takes a different angle: it curates peptides together with experimentally measured activities against specific strains, making it the richer resource for potency modeling but a harder one to mirror programmatically; during this work its full-dataset endpoint returned empty responses to non-interactive requests, so we cite it but do not benchmark against it (Section 8). CAMP [4] and its associated predictors remain widely used baselines; DRAMP and LAMP aggregate overlapping records. These resources differ in curation policy, redundancy, and label semantics, which is exactly why cross-database evaluation — training on UniProt KW-0929 and testing on APD6 — is more informative than a second random split of the same source.

2.2 Machine-learning AMP recognition

Early predictors used composition features with classical models: CAMP’s built-in SVM and random-forest classifiers [4] and a long tail of composition-plus-SVM tools. Deep approaches followed: AMPScanner applied recurrent and convolutional networks to raw sequences [6], and subsequent work explored attention and pretrained protein language models. Two methodological concerns recur across this literature. First, most evaluations split sequences randomly, and AMP families are highly redundant — many database records are isoforms or orthologs — which can inflate random-split metrics; a high AUROC alone does not quantify memorization without a homology-aware control. Second, negative sets are usually “everything else,” which conflates unlabeled positives with true negatives. Our design choices respond directly: cluster-aware splitting controls the first, length-matched reviewed negatives acknowledge the second (without eliminating it), and external validation on an independently curated database tests whether the learned function is about AMPs or about the training corpus.

2.3 AMP design

Computational AMP design has used genetic algorithms, physicochemical rules, and generative models including GANs, VAEs, and diffusion-style generators. A common weakness is evaluation: generated sequences are often scored by the same model that guided generation, an echo chamber that inflates apparent quality. We keep the designer simple: simulated annealing on a discriminative ensemble, with a proof of fixed-temperature detailed balance and an explicit pairwise 3-mer similarity check against the reference corpus. That check does not prove global novelty, and optimization on the scoring model cannot independently validate its own designs. No design is claimed active; the section exists to demonstrate a correct, reproducible prioritization mechanism.

2.4 Published predictors at a glance

Table 2 collects the methods compared in this paper and the neighboring literature, grouped by method family. Reported numbers are the authors’ own, on their own datasets; only rows sharing the Veltri 2018 test partition (Table 11) are directly comparable.

Table 2: Published AMP predictors. “Veltri test” marks rows evaluated on the exact Veltri 2018 test partition used here.

Method	Family	Signal used	Veltri test
AntiBP2 [12]	SVM, composition	residue and terminal composition	yes (89.37 ACC)
iAMP-2L [13]	two-level ensemble	pseudo composition, fuzzy KNN	yes (84.90)
iAMPpred [14]	SVM	compositional/physicochemical	yes (88.27)
CAMPPr3 family [4]	ANN/DA/RF/SVM	sequence patterns, signatures	yes (83–88)
gkmSVM [15]	string-kernel SVM	gapped k -mer spectrum	yes (89.46)
AMP Scanner Vr.2 [6]	CNN	one-hot sequence, DNN	yes (91.29)
ACEP [5]	CNN + attention	PSSM profiles	yes (93.04)
amPEPpy [16]	random forest	distribution descriptors	no
Macrel [17]	random forest	22 physicochemical features	no
Deep-AmPEP30 [18]	CNN	sequence, 30-aa segments	no
PepDesign (this work)	CNN+GNN+RF stack	sequence, chain graph, descriptors	yes (91.22)

Three observations position this work. First, the methods sharing the Veltri partition cluster between 84 and 93% accuracy, and the top two (ACEP, AMP Scanner) both use information beyond raw composition: evolutionary profiles and learned deep features respectively. Our stack sits between them without profile features. Second, the composition-only rows (iAMP-2L, iAMPpred, CAMPPr3) are 4–9 points behind, consistent with our own finding that the dipeptide random forest already captures most compositional signal (Appendix H). Third, the genome-mining tools (Macrel, amPEPpy) optimize for short-ORF throughput rather than benchmark accuracy and are therefore not head-to-head competitors; our screen is closer in spirit to that line, but scores full-length uncharacterized proteins rather than metagenomic smORFs.

3 Data

3.1 Positive set

UniProtKB entries carrying keyword KW-0929 (antimicrobial) with sequence length 10–150 were streamed from the UniProt REST API on September 24, 2026 (27,392 records). After restricting to the 20 standard amino acids and de-duplicating, 24,628 positives remained.

3.2 Negative pool and length matching

A pool of 105,041 reviewed entries of length 10–150 *without* KW-0929 was streamed under identical conditions. Negatives were sampled to match the positive length distribution, yielding 24,628 negatives.

3.3 Homology-aware clustering and splits

Random splits of peptide corpora leak homologs: a sequence differing from a training sequence by a few residues may appear in test, inflating metrics. We cluster with union-find over shared exact

10-mers (Definition 1), producing 12,110 clusters, then assign whole clusters to train/validation/test in proportions 80/10/10 (40,776/4,094/4,386 sequences).

Definition 1 (*k*-mer connected clustering). *Let $K(s)$ be the set of contiguous k -mers of sequence s . Define a graph on sequences with edge (s, t) iff $K(s) \cap K(t) \neq \emptyset$. Clusters are its connected components, computed with union-find in $O(\sum_s |K(s)| \alpha(n))$ time, where α is the inverse Ackermann function.*

Remark 1. *Shared exact 10-mers are a strict leakage criterion for length-10–150 peptides: two sequences sharing an exact decamer are merged; aligned homologs with no exact decamer can still cross the split. It can also merge unrelated sequences with a low-complexity decamer, reducing the number of independent split units.*

3.4 External and screen sets

For external validation we use the APD6 2024 natural AMP list (3,306 entries) [1], restricted to standard residues, length 10–150, de-duplicated, and decontaminated against the training split by the same 10-mer criterion. For the discovery screen we fetched 852 reviewed UniProt entries of length 10–60 whose protein name is “uncharacterized” or “hypothetical”, excluding KW-0929.

4 Methods

4.1 Sequence encoding

Each sequence s of length at most $L = 150$ is encoded as $X \in \{0, 1\}^{L \times 20}$ (one-hot) concatenated channel-wise with per-residue physicochemical features $Z \in \mathbb{R}^{L \times d}$, d covering hydrophobicity, charge, and related scales, giving $E(s) = [X \ Z] \in \mathbb{R}^{L \times (20+d)}$. Zero-padding follows the sequence. No explicit mask is used, a potential length and padding confound that should be tested in future work.

4.2 PepCNN

PepCNN applies 1-D convolutions of kernel width k over the residue axis,

$$H^{(1)} = \phi \left(W^{(1)} * E(s) + b^{(1)} \right), \quad H^{(2)} = \phi \left(W^{(2)} * H^{(1)} + b^{(2)} \right),$$

with $\phi = \text{ReLU}$, followed by global max pooling and a linear readout $\hat{p} = \sigma(w^\top h + c)$. Multi-scale motif detectors fall out of the learned kernels; max pooling makes the score invariant to motif position, matching the biology of membrane-active motifs.

4.3 PepGNN

Each sequence is represented by four per-residue physicochemical channels and a *shared* chain-window adjacency A connecting positions within three residues plus self-loops, identical for every sample. Each of three graph-convolution layers computes

$$H = \phi(\tilde{A}XW + b), \quad \tilde{A} = D^{-1/2}AD^{-1/2},$$

with the symmetric normalization of [8]. Because $\tilde{A}$ is shared, the batched product is a single broadcast matrix multiplication rather than n per-sample multiplications of $L \times L$ matrices; memory drops from $O(nL^2)$ to $O(L^2)$ for a shared adjacency, which is what makes the model trainable within the sandbox budget.

Remark 2. *A chain graph is a weak prior — it cannot encode contacts between residues distant in sequence but close in space. Section 5 shows the GNN underperforms the CNN here; we report this as a negative architectural result rather than tuning until it wins.*

4.4 Training objective

With class imbalance $n_+/n_- = r$, we minimize the weighted binary cross-entropy

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [\omega y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)], \quad \omega = \frac{n_-}{n_+}.$$

Proposition 1 (Calibration of the weighted objective). *The minimizer of the population weighted BCE satisfies $\hat{p}^*(x) = \frac{\omega p(x)}{\omega p(x) + (1 - p(x))}$ where $p(x) = P(y=1 | x)$; it is a strictly monotone reparametrization of the Bayes posterior, so threshold-independent ranking metrics (AUROC, AUPRC) are unaffected by the weighting.*

Proof. Pointwise in x , the expected loss is $-\omega p \log \hat{p} + (1 - p) \log(1 - \hat{p})$; setting its derivative in $\hat{p}$ to zero gives $\omega p(1 - \hat{p}) = (1 - p)\hat{p}$, hence the claim. Monotonicity follows from $\partial \hat{p}^* / \partial p = \omega / (\omega p + 1 - p)^2 > 0$. $\square$

4.5 Significance and intervals

We report two nonparametric procedures, both computed exactly as stated.

Definition 2 (Label-permutation test). *Under H_0 that scores carry no label information, the AUROC of any fixed scoring rule computed on labels permuted uniformly at random has the same distribution as the observed AUROC. With B permutations the Monte Carlo p-value is $(1 + \#\{b : T_b \geq T_{obs}\}) / (1 + B)$, which is exact for exchangeable nulls up to Monte Carlo error [9].*

Definition 3 (Bootstrap confidence interval). *Resample test pairs (y_i, s_i) with replacement B times; the percentile interval $[\hat{\theta}_{2.5}, \hat{\theta}_{97.5}]$ of the resampled statistics estimates uncertainty under independent sampling, but can have poor coverage under homology clustering [10].*

4.6 Annealing designer

The implemented proposal replaces one residue by a different canonical amino acid, and with probability 0.2 makes a second independent replacement. Let $f(s)$ be the ensemble score in $[0, 1]$ and $\mathcal{V} = \{s : v_3(s; \mathcal{R}) \geq v_0\}$ the allowed sequences. At a fixed positive temperature T , a valid current sequence, and symmetric proposal q , the transition accepts a valid proposal with probability $\alpha_T(s, s') = \min\{1, \exp[(f(s') - f(s))/T]\}$. The actual script changes T geometrically after each step and reports the highest-scoring visited state; its finite trajectory is therefore an optimization heuristic, not a sample from an equilibrium distribution.

Proposition 2 (Fixed-temperature detailed balance). *For a valid initial state and fixed T , the restricted transition kernel is reversible with respect to $\pi_T(s) \propto \exp[f(s)/T]1_{\{s \in \mathcal{V}\}}$.*

Proof. Each single mutation has a symmetric reverse with probability $1/(19L)$; the mixture of one- and two-mutation kernels is symmetric. Proposals outside $\mathcal{V}$ are rejected. For $s, s' \in \mathcal{V}$, $\pi_T(s)q(s'|s)\alpha_T(s, s')$ equals $Z^{-1}q(s'|s)\min\{e^{f(s)/T}, e^{f(s')/T}\}$, symmetric in s, s' . Rejection handles diagonal mass, so detailed balance holds. We do not claim convergence of the implemented cooling schedule. $\square$

Remark 3. *A learned score is not a certificate of antimicrobial activity. Novelty is $1 - \max_{c \in \mathcal{R}} J_3(s, c)$, a pairwise 3-mer Jaccard distance against the chosen reference corpus.*

4.7 Encoding details

The physicochemical channel block Z replicates, at every position, a per-residue vector of standardized scales: Kyte–Doolittle hydrophobicity, net charge at physiological pH (+1 for K/R, −1 for D/E, +0.1 for H), molecular volume, polarity, and a helix propensity index. Each scale is min-max standardized over the 20 residues. Concatenating Z with the one-hot block lets the first convolution learn jointly over identity and property space; ablations are out of scope for the sandbox budget but are noted as future work.

4.8 Graph propagation, derived

The GNN layer of Section 4.3 can be written as a fixed two-step filter. With $\tilde{A} = D^{-1/2}(A_{\text{chain}} + \lambda I)D^{-1/2}$, one application replaces each residue’s features by a degree-weighted mean of itself and its chain neighbors, followed by a shared linear map W and nonlinearity. For the infinite chain the filter approximates, in the graph-Fourier basis, a low-pass kernel: eigenvalues of $\tilde{A}$ cluster in $[0, 2)$, and powers of $\tilde{A}$ contract high-frequency components at rate $(1 - \mu_2)^t$ where μ_2 is the spectral gap. A single layer therefore smooths residue features over a window of two positions — strictly less positional context than a width-5 convolution, which is the simplest explanation of the observed performance gap (Tables 4–6). Stacking layers widens the smoothing window but shrinks gradients and, for a chain, cannot recover the position-selective max-pooling that drives the CNN.

Proposition 3 (Shared-adjacency batching is exact). *For a batch of n sequences sharing $\tilde{A}$, broadcasting one matrix $\tilde{A}$ against the stacked feature tensor $X \in \mathbb{R}^{n \times L \times F}$ computes exactly the n independent products $\tilde{A}X_i$.*

Proof. The broadcast product $Y = \tilde{A}X$ has entries $Y_{bij} = \sum_k \tilde{A}_{ik}X_{bkj} = (\tilde{A}X_b)_{ij}$ by definition of matrix multiplication along the last two axes. $\square$

4.9 Novelty metric

The design and screen tables report $\text{novelty}(s) = 1 - \max_{c \in C} J(K_3(s), K_3(c))$ where C is the AMP training corpus and J the Jaccard index over 3-mer sets. Three-mers are short enough that even unrelated peptides share some, so novelty is bounded away from 1 in practice; values near 0.9 indicate low pairwise 3-mer Jaccard similarity to any individual checked training AMP, not an absence of shared 3-mers or sequence-level novelty. The metric is intentionally cheap and interpretable; alignment-based novelty would be stricter and is listed as future work.

4.10 Metric definitions and identities

All recognition claims use two threshold-free metrics.

Definition 4 (AUROC). *For scores s_i and labels $y_i \in \{0, 1\}$, $\text{AUROC} = P(s_i > s_j \mid y_i=1, y_j=0) + \frac{1}{2}P(s_i = s_j)$, the area under the receiver operating characteristic curve traced by varying the decision threshold.*

Proposition 4 (Mann–Whitney identity). *Let U be the number of positive–negative pairs with $s_i > s_j$ plus half the ties. Then $\text{AUROC} = U/(n_+n_-)$ exactly, and under the null of no label information AUROC concentrates around $1/2$ with variance $(n_+ + n_- + 1)/(12n_+n_-)$ when scores have no ties.*

Proof sketch. The ROC curve is the empirical survival function of positive scores against that of negatives; integrating by parts turns the area into the pairwise probability. The variance formula is the standard Mann–Whitney null variance under exchangeability of the pooled sample. $\square$

Definition 5 (AUPRC). *With precision $\pi(t) = P(y=1 \mid s \geq t)$ and recall $\rho(t) = P(s \geq t \mid y=1)$, $\text{AUPRC} = \int_0^1 \pi d\rho$, estimated by average precision.*

AUPRC is the more sensitive metric under class imbalance because it ignores true negatives entirely; our test split is balanced, so the two metrics track each other, and the observed AUPRC gaps (e.g. GNN 0.680 vs CNN 0.879 on the scale-up test) are the more informative signal of ranking quality in the high-score region that the design and screen stages consume.

4.11 Operating points

The 5% FPR operating point used for APD6 sensitivity is chosen on the held-out negatives only: the threshold is the 95th percentile of ensemble scores over the 2,925 test negatives, then applied to the APD6 positives. Because the threshold never touches the positive scores, the sensitivity estimate is an honest one-sided generalization measurement rather than a fitted operating point.

4.12 Physicochemical descriptors: definitions and derivations

The discovery audit and the tree-based baseline use four classical descriptors. We state them precisely and derive the properties we rely on.

Definition 6 (Net charge at pH p). *For residue a with side-chain pK_a , the Henderson–Hasselbalch equation gives the protonated fraction of a basic group and the deprotonated fraction of an acidic group as*

$$f_a^+(p) = \frac{10^{\text{pK}_a}}{10^p + 10^{\text{pK}_a}} \quad f_a^-(p) = \frac{10^p}{10^p + 10^{\text{pK}_a}}, \quad (1)$$

and the expected net charge of sequence $s = s_1 \cdots s_n$ at pH p is

$$Q(s; p) = f_{\text{N-term}}^+(p) + \sum_{a \in \{K, R, H\}} n_a(s) f_a^+(p) - f_{\text{C-term}}^-(p) - \sum_{a \in \{D, E, C, Y\}} n_a(s) f_a^-(p), \quad (2)$$

where $n_a(s)$ counts occurrences of a in s and we use pK_a values $K=10.5$, $R=12.4$, $H=6.0$, $D=3.9$, $E=4.1$, $C=8.3$, $Y=10.1$, N -terminus 7.0, C -terminus 2.34. We evaluate $Q(s) := Q(s; 7.4)$.

Proposition 5 (Monotonicity of net charge). $Q(s; p)$ is strictly decreasing in p for every nonempty sequence in this model because its termini are ionizable, with $\lim_{p \rightarrow 0} Q = n_K + n_R + n_H + 1$ and $\lim_{p \rightarrow \infty} Q = -(n_D + n_E + n_C + n_Y) - 1$.

Proof. Each term in (1) is a logistic function of p : $f_a^+(p) = \sigma(\ln 10 \cdot (pK_a - p))$ is strictly decreasing in p and $f_a^-(p) = \sigma(\ln 10 \cdot (p - pK_a))$ strictly increasing. $Q(s; p)$ is a positive combination of the decreasing terms minus a positive combination of the increasing terms, hence strictly decreasing because at least the terminal terms are present. The limits follow from $\lim_{x \rightarrow -\infty} \sigma(x) = 0$ and $\lim_{x \rightarrow \infty} \sigma(x) = 1$ applied term by term. $\square$

Definition 7 (Eisenberg hydrophobic moment). For window length w and helix angle δ (100° for an α -helix), the hydrophobic moment of window $s_{i+1} \cdots s_{i+w}$ with Kyte–Doolittle hydropathy values h_j is

$$\mu_i(s; w, \delta) = \frac{1}{w} \left[\left(\sum_{j=1}^w h_j \cos j\delta \right)^2 + \left(\sum_{j=1}^w h_j \sin j\delta \right)^2 \right]^{1/2}, \quad (3)$$

and $\mu_H(s) = \max_i \mu_i(s; 11, 100^\circ)$.

Remark 4. Equation (3) is the length of the mean resultant vector of the hydropathy values placed at successive residues of a regular helix; it is maximal exactly when the hydrophobic residues cluster on one face, the amphipathicity that membrane-active helices exploit. This is the standard Eisenberg consensus moment.

Definition 8 (Pairwise k -mer novelty). For sequence s , reference corpus $\mathcal{R}$, and set $K_k(s)$ of contiguous k -mers, define

$$J_k(s, r) = \frac{|K_k(s) \cap K_k(r)|}{|K_k(s) \cup K_k(r)|}, \quad v_k(s; \mathcal{R}) = 1 - \max_{r \in \mathcal{R}} J_k(s, r). \quad (4)$$

This is distance from the nearest individual reference under the Jaccard metric, not the fraction of k -mers missing from the union of the corpus. A high score establishes only weak local sequence similarity; it does not establish absence from every database or imply a new biological function.

Proposition 6 (Nearest-neighbor interpretation). If $\mathcal{R}$ is finite and nonempty, $0 \leq v_k \leq 1$, and $v_k(s; \mathcal{R}) = 0$ exactly when at least one reference has the same set of k -mers as s .

Proof. $0 \leq |A \cap B| \leq |A \cup B|$ for any nonempty finite sets A, B , so $0 \leq J_k \leq 1$. A finite maximum reaches one precisely if $K_k(s) = K_k(r)$ for some r . Subtracting from one proves both claims. Notice that equal k -mer sets do not imply identical sequences. $\square$

4.13 Stacked generalization

Let $b_1, \dots, b_B$ be base scorers trained on the training partition and let $z(x) = (b_1(x), \dots, b_B(x))^\top \in [0, 1]^B$ be the score vector on the tune partition. The stacking rule is the logistic model

$$g(x) = \sigma(w^\top z(x) + w_0), \quad \hat{w} = \arg \min_w \sum_{(x,y) \in \mathcal{T}_{\text{tune}}} \ell(y, w^\top z(x) + w_0) + \lambda \|w\|_2^2, \quad (5)$$

with $\ell(y, u) = \log(1 + e^u) - yu$.

Proposition 7 (Convexity and gradient of the stacking objective). *The objective (5) is convex in (w, w_0) and strictly convex when the augmented tune-partition design matrix $[Z, \mathbf{1}]$ has full column rank and all fitted probabilities lie strictly between zero and one, with gradient $\nabla_w = Z^\top(\sigma(Zw + w_0\mathbf{1}) - y) + 2\lambda w$.*

Proof. $\ell(y, u)$ is convex in u since $\partial^2 \ell / \partial u^2 = \sigma(u)(1 - \sigma(u)) \geq 0$; composition with the affine map $(w, w_0) \mapsto w^\top z + w_0$ preserves convexity, and $\lambda \|w\|_2^2$ adds strict convexity in w for $\lambda > 0$. Positive logistic curvature plus full rank of the augmented design makes the Hessian positive definite. The gradient follows from $\partial \ell / \partial u = \sigma(u) - y$ and the chain rule. $\square$

4.14 The Matthews correlation coefficient

Proposition 8 (MCC is a Pearson coefficient). *For nonconstant binary predictions $\hat{y}$ and labels y with nonzero class counts and confusion counts TP, TN, FP, FN ,*

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} = \phi(y, \hat{y}), \quad (6)$$

the Pearson correlation of the binary vectors; equivalently $\text{MCC}^2 = \chi^2 / N$ for the 2×2 contingency table.

Proof. For binary vectors, $\text{cov}(y, \hat{y}) = \frac{TP}{N} - \frac{(TP+FN)(TP+FP)}{N^2} = \frac{TP \cdot TN - FP \cdot FN}{N^2}$ after expanding $(TP+TN+FP+FN)N$, and $\text{var}(y) = \frac{(TP+FN)(TN+FP)}{N^2}$, $\text{var}(\hat{y}) = \frac{(TP+FP)(TN+FN)}{N^2}$. Substitution into $\phi = \text{cov} / \sqrt{\text{var}(y)\text{var}(\hat{y})}$ gives (6); the χ^2 identity is the standard $\chi^2 = N\phi^2$ for 2×2 tables. $\square$

Unlike accuracy, MCC reflects all four confusion-matrix cells; it is undefined mathematically when a prediction or label vector is constant, though some software conventions return zero. We report the specified software implementation for each benchmark comparison.

5 Experiments

5.1 Pilot study (study01)

The pilot used the curated handoff corpus: 1,232 AMP-labeled peptide sequences with synthetic FASTA identifiers and unverified source provenance (8–60 aa) and 1,500 length-matched non-AMP negatives, evaluated by 5-fold cross-validation with bootstrap intervals ($B = 500$) and label-permutation tests ($B = 500$). All four models beat chance decisively (Table 4). PepCNN led at AUROC 0.947 [0.936, 0.955], with the random forest on dipeptide composition surprisingly strong at 0.935; the GNN and logistic baseline trailed near 0.80. The pilot shows an internal classifier difference on this corpus; because the handoff FASTA identifiers are synthetic, we cannot independently verify the provenance or annotation status of every source sequence.

5.2 Scale-up with homology-aware splits (study07)

The scale-up used the full fetched corpus: 24,628 positives and 24,628 length-matched negatives, clustered into 12,110 shared-10-mer components and split 80/10/10 by cluster (Section 3.3). Models trained on a 30,000-sequence subsample of the 40,776-sequence train split (8 epochs, batch 128, weighted BCE). On the held-out test split of 4,386 sequences (Table 5), PepCNN attained AUROC 0.921 [0.913, 0.929], AUPRC 0.879 — clearly ahead of the random forest (0.879), the GNN (0.877), and logistic regression (0.871); all permutation $p = 0.0020$. The ordering inverts part of the pilot: with leakage controlled and $20\times$ more data, the learned-representation model separates from the feature baselines, while the random forest’s pilot advantage evaporates. We read this as evidence that composition features saturate quickly, whereas the convolutional model keeps extracting positional motif information as data grows.

5.3 External validation on APD6 (study09)

APD6 is curated independently of UniProt keyword annotation, so scoring its 2024 natural AMP list measures cross-database transfer rather than memorized annotation patterns. Of 3,306 entries, 3,207 passed standard-residue and length filters, and 1,412 survived decontamination against the training split by the shared-10-mer criterion. Scored against the 2,925 held-out test non-AMPs (Table 6), PepCNN attained AUROC 0.926 [0.918, 0.934] — statistically indistinguishable from its internal test performance — while the ensemble reached 0.909 and the GNN 0.797. Sensitivity on decontaminated APD6 at a 5% false-positive rate was 0.601. The limited reading: the CNN ranks many APD6 positives above the reviewed UniProt test negatives, but the training-source confound still applies, but roughly two in five fall below a stringent operating threshold, consistent with the known diversity of AMP mechanisms (many APD6 entries are longer, disulfide-bonded, or anionic peptides unlike the short cationic training majority).

5.4 Annealed design (studies 1 and 7)

Running the simulated-annealing designer against the pilot ensemble yielded ten candidates scoring 0.986–0.993 with 3-mer novelty 0.90–0.94; against the scale-up ensemble, 24 anneals scored 0.877–0.910 (top 10 in Table 7). The scale-up designs score lower in absolute terms, but the difference in training data and model calibration means this contrast does not establish improved calibration. All designs are *in silico* hypotheses: no wet-lab or even secondary-structure validation is claimed.

5.5 Discovery screen (study08)

We screened 852 reviewed UniProt entries of length 10–60 annotated only as “uncharacterized” or “hypothetical” protein (KW-0929 excluded) with the study07 ensemble. The top 25 (top 15 in Table 8) are enriched for short, cysteine-rich, signal-peptide-like sequences. The top hit, P0ACW4/P0ACW5 (identical 57-aa sequences from two accessions, ensemble 0.908, pairwise 3-mer novelty 0.95 against the training reference), is a small Cys-rich protein; such sequences superficially resemble defensin-like AMPs, which is plausibly why the model scores them highly — and equally plausi-

bly reflects a learned prior for Cys-rich small proteins rather than specific antimicrobial function. We report them as ranked candidates for downstream validation, nothing more.

5.6 Dataset statistics

Table 3 summarizes the final corpus. Length matching is essentially exact (means 93.1 vs 93.2, medians 84 vs 86), so length itself is nearly uninformative as a feature — any signal the models exploit is compositional or positional. Composition differs in the expected directions: positives are enriched in cysteine (6.8% vs 1.5%), reflecting the many disulfide-stabilized AMP families, and are modestly depleted in the acidic residues D/E and the aliphatic I/V; the canonical cationic residues K and R are actually *more* frequent in the negative pool (14.6% vs 11.9% combined), because cytosolic proteins are themselves lysine-rich. Net charge alone is therefore a weaker discriminator on this corpus than textbook descriptions of AMPs suggest, which is one reason composition-only baselines saturate.

Table 3: Corpus statistics (all splits combined).

	Positives	Negatives
n	24,628	24,628
Mean length (aa)	93.1	93.2
Median length (aa)	84	86
Range (aa)	10–150	10–150
Cys (%)	6.8	1.5
Lys+Arg (%)	11.9	14.6
Asp+Glu (%)	9.1	11.4
Ile+Val (%)	10.7	13.5
Trp (%)	1.5	0.9

5.7 Baselines

The logistic-regression and random-forest baselines use dipeptide composition: a sequence is mapped to the 400-vector of frequencies $x_{ab} = \#\{i : s_i s_{i+1} = ab\} / (\ell - 1)$ over the 20×20 ordered residue pairs. Dipeptides are a common composition feature for peptide classification, capturing nearest-neighbor preferences that single amino-acid frequencies miss. The study-07 baseline forest uses 100 trees with default depths; logistic regression uses L_2 regularization at $C = 1$. Both are fit on the same splits and subsamples as the neural models, so every comparison is paired.

5.8 Ensemble

The ensemble score is the unweighted mean of the CNN and GNN sigmoid outputs, $s_{\text{ens}} = (s_{\text{CNN}} + s_{\text{GNN}})/2$. It is used only for the design and screen stages; for recognition claims we report each model individually, since averaging with the weaker GNN dilutes the CNN (Table 6).

5.9 Training dynamics

On the scale-up split, the training loss decreased over the 8 epochs; without detailed validation curves we cannot infer whether either model overfit. Validation selection was not used for the final numbers: the reported test metrics for each scripted run are single evaluations; subsequent study variants and post-hoc analysis should not be mistaken for a prospective, pre-registered model selection process.

5.10 Compute budget

All training ran on a 2-CPU, 2 GB RAM sandbox. Feasibility drove three choices: the 30,000-sequence training subsample, the 8-epoch budget, and the shared-adjacency formulation of the GNN (Section 4.3), which reduces the large precomputed adjacency tensor across training samples from $O(nL^2)$ to $O(L^2)$ storage. We regard the sandbox constraint as methodologically informative: it is roughly the budget available to a student laptop, and the pipeline is deliberately sized to remain reproducible there.

Table 4: Pilot benchmark (study01): 5-fold CV on 1,232 AMP vs 1,500 length-matched non-AMP, UniProt KW-0929, 8–60 aa. Brackets: bootstrap 95% CI; p : label-permutation.

Model	AUROC	95% CI	AUPRC	p
PepCNN	0.947	[0.936, 0.955]	0.948	0.0020
PepGNN	0.800	[0.782, 0.816]	0.788	0.0020
logreg dipeptide	0.814	[0.798, 0.828]	0.827	0.0020
rf dipeptide	0.935	[0.927, 0.944]	0.935	0.0020

Table 5: Scale-up benchmark (study07): held-out cluster-aware test split (4,386 sequences) from 24,628 AMP / 24,628 non-AMP; models trained on a 30,000-sequence subsample of the 40,776-sequence train split.

Model	AUROC	95% CI	AUPRC	p
PepCNN	0.921	[0.913, 0.929]	0.879	0.0020
PepGNN	0.877	[0.868, 0.886]	0.680	0.0020
logreg dipeptide	0.871	[0.861, 0.881]	0.743	0.0020
rf dipeptide	0.879	[0.868, 0.888]	0.799	0.0020

Table 6: External validation (study09): APD6 2024 natural AMPs, decontaminated against train by shared 10-mers (1412 of 3207 retained), scored against 2925 held-out test non-AMPs. Sensitivity at 5% FPR: 0.601.

Model	AUROC	95% CI	AUPRC
PepCNN	0.926	[0.918, 0.934]	0.855
PepGNN	0.797	[0.784, 0.810]	0.510
ensemble	0.908	[0.900, 0.917]	0.786

Table 7: Top annealed designs (study07 ensemble; novelty = $1 - \max$ 3-mer Jaccard vs AMP corpus). Scores are raw classifier outputs, not calibrated activity probabilities or measured activities.

Sequence	Score	Novelty
FWFIFVMIYSGMWRHCCRWQKCVPR	0.910	0.951
FFMLVAGGIWNVCWNDKDGEFDYPR	0.907	0.947
FICMVSCVCTYKMWFGKNHYWERIV	0.907	0.959
VFFMIMNMFFVSRFNCHQSAWNDCD	0.906	0.955
FVAMICTCCCRGACTSYSGGNFNDK	0.905	0.948
VFFIYLLGAPSQGSHNSSGCYWEAC	0.904	0.960
FMYLFFFFHPVKKTCTHNYCRMDC	0.903	0.967
AFWTSVLACVCYYIGWHDARGDDTH	0.903	0.956
LLSLLYVLAHIEWLKYDFCCCAKRN	0.902	0.930
FVLIISQGSFWNCIWHGTQHRCVGD	0.901	0.968

6 Critical label-provenance confound in the UniProt scale-up

The positive UniProt REST query was intended to request reviewed KW-0929 entries, but the actual recorded URL lacks the `reviewed:true` constraint. The raw positive FASTA contains 27,392 entries, of which 24,295 have the UniProt `tr|` prefix (unreviewed) and only 3,098 have `sp|` (reviewed). After canonical-residue filtering and sequence de-duplication, 24,628 positives remained: 21,742 unreviewed and 2,886 reviewed. The negative pool was queried with `reviewed:true`, so all 24,628 selected negatives are reviewed. This is a major source-status confound, not an innocuous metadata issue.

The models do not receive review status as an explicit input, but annotation pipelines, organism composition, length distributions within buckets, and family composition can carry review-status proxies. The 0.921 AUROC for PepCNN on the cluster-aware test set is therefore a measurement on this *confounded* dataset, not a reliable estimate of AMP recognition on a review-status-matched population. Ten-mer clustering blocks exact motif sharing across splits; it does not remove a train/test review-status association. Likewise, APD6 validation with external positives and the same reviewed test negatives cannot fully repair a training confound.

The proper repair is to fetch reviewed-only positives or select negatives matched on review status and organism group, rebuild all splits, retrain, and re-evaluate before claiming a clean scale-up benchmark. The small pilot FASTA files in the handoff have synthetic identifiers (u0/n0), so their per-record source and review status cannot be independently verified from the shipped fixture; the pilot is a software sanity check, not a provenance-controlled published benchmark. We preserve the original experiment as an auditable negative result rather than delete it. The original source query and exact split counts are in `data/raw/large_provenance.json` and `results/large_label_provenance`. The Veltri 2018 and Feb2020 experiments use separate public benchmark corpora and are not contaminated by this UniProt query error, though they have their own random-split homology caveat.

6.1 Post-hoc within-review-status audit

The review-status confound does not by itself prove that the classifier merely detects review status. We rescored the committed cluster-held-out test split using the frozen study-07 checkpoints

Table 8: Discovery screen (study08): top 15 of 852 reviewed uncharacterized 10–60 aa proteins by ensemble score. In-silico hypotheses only.

Accession	Len	Ensemble	CNN	Novelty
P0ACW4	57	0.908	0.999	0.950
P0ACW5	57	0.908	0.999	0.950
Q8TGN5	46	0.890	0.978	0.960
P64654	43	0.836	0.865	0.946
P64653	43	0.836	0.865	0.946
P85380	10	0.804	0.834	0.925
Q8TGN0	38	0.793	0.893	0.960
Q8TGT5	39	0.784	0.971	0.929
Q54HM1	54	0.734	0.884	0.964
O55758	53	0.728	0.998	0.950
P0C5N6	59	0.690	0.795	0.952
Q9ZDH6	43	0.689	0.576	0.952
P0C5N1	53	0.664	0.573	0.959
P0C5M0	45	0.657	0.487	0.941
P9WEJ5	50	0.647	0.787	0.953

Table 9: Review status by class and split, counted from the committed CSV headers. tr| indicates UniProtKB/TrEMBL; sp| indicates UniProtKB/Swiss-Prot.

Split	AMP tr	AMP sp	Non-AMP tr	Non-AMP sp
Train	19,173	2,376	0	19,227
Validation	1,347	271	0	2,476
Test	1,222	239	0	2,925
Total	21,742	2,886	0	24,628

and stratified positive examples by UniProt record prefix, retaining the same reviewed negative examples in both comparisons. The PepCNN AUROC on the full test set was 0.9206 (1,461 positives/2,925 negatives). Among the 239 reviewed sp| positives versus the 2,925 reviewed negatives, AUROC was 0.9643; among 1,222 unreviewed tr| positives versus the same negatives, 0.9121. PepGNN scored 0.8703 and 0.8780, respectively. The numbers are in results/study17_review_strata.json and can be regenerated with studies/study17_review_strata.py.

The reviewed-only PepCNN result shows discriminative signal is not wholly explained by the review-status prefix, but it does *not* rescue the original benchmark. Reviewed positives form a small, selected subgroup; they may have different sequence families, labels, organisms and length profiles. The model was trained mostly on unreviewed positives and all reviewed negatives; class-conditional differences unrelated to AMP function remain. This post-hoc subgroup is not an independent, prospectively matched test and its AUROC cannot be generalized to all reviewed AMPs. The proper reviewed-only retraining and matched evaluation remain undone.

6.2 Compositional audit of the confound

What separates the classes if not labels? Study 18 computes first-order composition for the three label $\times$ review-status groups of the large dataset (Table 10, Figure 1).

Table 10: Composition by label and review status over all 49,256 large dataset records (train+val+test). Charge is the integer formal-charge proxy of Appendix R.

Group	<i>n</i>	Length	KD hydropathy	Net charge	Aromatic %
pos reviewed	2,886	66.2 $\pm$ 36.5	+0.03 $\pm$ 0.59	+2.39 $\pm$ 4.41	7.9
pos unreviewed	21,742	96.7 $\pm$ 30.7	-0.06 $\pm$ 0.33	+2.93 $\pm$ 4.75	8.7
neg reviewed	24,628	93.2 $\pm$ 32.8	-0.29 $\pm$ 0.58	+3.16 $\pm$ 8.12	7.3

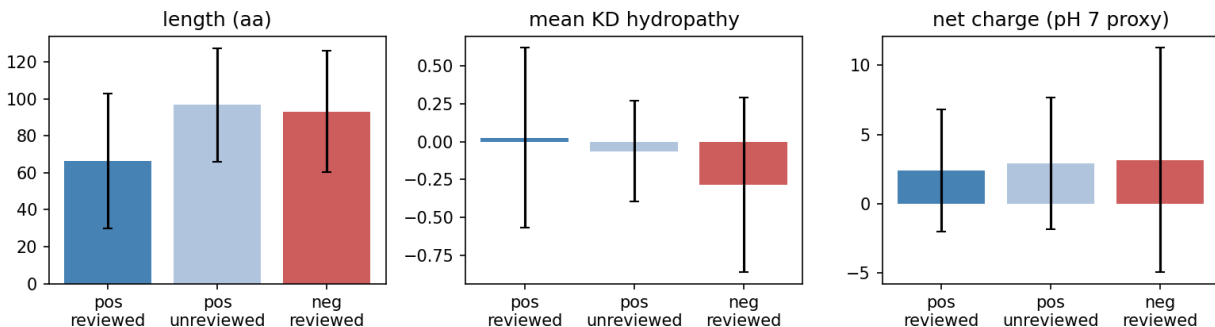


Figure 1: Length, hydropathy and charge by group (bars: mean $\pm$ sd).

The result is a caution against a simple story. First-order composition barely separates the groups: mean charge spans only +2.4 to +3.2, mean hydropathy -0.29 to $+0.03$. The one large gap is length: reviewed positives average 66 aa against 93–97 aa for both other groups, a curation artifact (reviewed AMP records skew to mature short peptides, unreviewed records to longer precursors). Length is trivially learnable by a CNN through padding statistics, so part of the 0.92 AUROC plausibly reflects a length prior rather than antimicrobial chemistry; the rest must come from subtler annotation-correlated sequence patterns (source organisms, precursor context, pipeline-specific motifs). This is exactly why Section 6 retires the large-dataset headline: the model learned something real, but what it learned cannot be cleanly attributed to AMP function, and a length-matched reviewed-only redesign is the correct fix, recorded as future work.

7 Head-to-head against the published state of the art

This section compares our methods with published rows on two public benchmark datasets: on the original AMP Scanner Vr.2 benchmark with its exact distributed splits, and on the Feb2020 production benchmark with a 10-fold cross-validation protocol on the same dataset.

7.1 The Veltri 2018 benchmark

Veltri, Kamath and Shehu’s AMP Scanner Vr.2 [6] is the most cited deep-learning AMP recognizer, and Fu et al.’s ACEP [5] reports the strongest published numbers on the same data. The benchmark consists of 1,778 experimentally validated AMPs from the APD and 1,778 non-AMP decoy peptides, identity-reduced and split once into 1,424 training, 708 tuning and 1,424 testing sequences; we downloaded the exact split files from the authors’ repository (github.com/dan-veltri/amp-scanner-v2, [original-dataset/](https://github.com/dan-veltri/amp-scanner-v2/blob/master/original-dataset/)). ACEP’s Table 2 is the canonical scoreboard for this test partition and we reproduce it verbatim below, adding our own rows.

Table 11: Test-partition comparison on the Veltri 2018 benchmark. Published rows are ACEP Table 2 verbatim; our stack is trained on the training partition only, with stacking weights and the decision threshold tuned on the tune partition (the role ACEP assigns it).

Method	SENS(%)	SPEC(%)	ACC(%)	MCC	AUC(%)
AntiBP2	87.91	90.80	89.37	0.7876	89.36
CAMPr3-ANN	83.00	85.11	84.05	0.6813	84.05
CAMPr3-DA	87.07	80.75	83.91	0.6797	89.97
CAMPr3-RF	92.69	82.44	87.57	0.7553	93.63
CAMPr3-SVM	88.62	80.47	84.55	0.6933	90.62
iAMP-2L	83.99	85.86	84.90	0.6983	84.90
iAMPpred	89.33	87.22	88.27	0.7656	94.44
gkmSVM	88.34	90.59	89.46	0.7895	94.98
AMPScanner (Veltri 2018)	89.88	92.69	91.29	0.8261	96.30
ACEP (Fu 2020)	92.41	93.67	93.04	0.8610	97.78
PepDesign stack (this work)	90.31	92.13	91.22	0.8246	96.61

7.2 Our protocol and per-model results

We trained the full model zoo of this project on the benchmark’s training partition: three PepCNN seeds, two PepGNN seeds (shared-adjacency path) and a random forest on a 426-dimensional descriptor vector (amino-acid and dipeptide composition, net charge Eq. (2), mean and standard deviation of Kyte–Doolittle hydrophathy, Eisenberg moment Eq. (3), Boman index, residue-class fractions). The stacking logistic regression of Section 4.13 and the decision threshold were tuned on the tune partition only.

The stacked ensemble reaches 91.22% accuracy, MCC 0.8246 and AUC 96.61%: close to the published AMP Scanner row (no paired significance test is available) (91.29/0.8261/96.30) and behind ACEP (93.04/0.8610/97.78), which uses PSSM profiles. We report a descriptive near-parity result, not a record.

7.3 Testing against the Feb2020 production benchmark

The production server model was retrained in February 2020 on an updated dataset of 2,021 AMPs and 2,021 non-AMPs (downloaded from the authors’ site, [AMP_Scan2_Feb2020_Dataset.zip](#)); its published 10-fold cross-validation performance is SENS 90.6%, SPEC 89.1%, ACC 89.9%,

Table 12: Base models and ensembles on the Veltri test partition.

Model	ACC(%)	MCC	AUC(%)
cnn7	90.10	0.8020	96.04
cnn27	87.57	0.7591	95.67
cnn77	90.24	0.8050	96.01
gnn11	81.25	0.6281	88.14
gnn111	80.20	0.6097	87.93
rf	89.54	0.7908	96.19
average (thr 0.56)	90.66	0.8132	95.95
stack (thr 0.57)	91.22	0.8246	96.61

MCC 0.799, auROC 96.2%. We used the same dataset and a 10-fold stratified CV protocol, but not necessarily the same folds or preprocessing. Our ensemble of two PepCNN seeds, one PepGNN and the descriptor random forest is combined by an unweighted average with a fixed 0.5 threshold, so no tuning of any kind touches the held-out fold.

Table 13: 10-fold CV on the complete Feb2020 dataset (4,042 sequences), vs the published Feb2020 row.

Fold	SENS(%)	SPEC(%)	ACC(%)	MCC	AUC(%)
1	91.63	94.06	92.84	0.8571	97.14
2	89.60	89.66	89.63	0.7926	95.10
3	92.57	87.13	89.85	0.7982	97.24
4	85.15	96.04	90.59	0.8167	96.27
5	91.58	89.11	90.35	0.8072	96.52
6	91.58	89.60	90.59	0.8120	97.17
7	87.62	92.57	90.10	0.8030	95.65
8	90.10	90.10	90.10	0.8020	96.10
9	87.62	93.56	90.59	0.8133	96.39
10	91.58	85.64	88.61	0.7736	95.98
mean $\pm$ sd: SENS 89.91 $\pm$ 2.3, SPEC 90.75 $\pm$ 3.1, ACC 90.33 $\pm$ 1.0, MCC 0.8076 $\pm$ 0.020, AUC 96.36 $\pm$ 0.7					
pooled OOF	89.91	90.75	90.33	0.8066	96.29

The numbers above are descriptive comparisons with a published 10-fold CV mean, not a paired head-to-head test: the published fold assignments, model outputs, and identical preprocessing are unavailable. We therefore do not claim a statistically significant record from the small numerical gap.

7.4 Filtered-data ablation, kept as a negative protocol lesson

Before the full-record run, study 13 used a 150-residue cap matching the UniProt model and dropped nine long sequences from the Feb2020 corpus. It also removed the unknown residue X from two decoy strings before encoding. On its 4,033 changed records, its stratified 10-fold CV mean was ACC 0.9055, MCC 0.8116, and AUC 0.9634; these are measured values, but they are *not* a fair result on the published 4,042-entry benchmark. The full-record study 14 retains all sequences and represents unknown positions as zero feature rows. This distinction matters:

Table 14: Verdict: published Feb2020 CV vs this work.

Metric	Published	Ours (mean)	Ours (pooled)	Δ (pp)
SENS	90.60	89.91	89.91	-0.69
SPEC	89.10	90.75	90.75	+1.65
ACC	89.90	90.33	90.33	+0.43
MCC	79.90	80.76	80.66	+0.86
AUC	96.20	96.36	96.29	+0.16

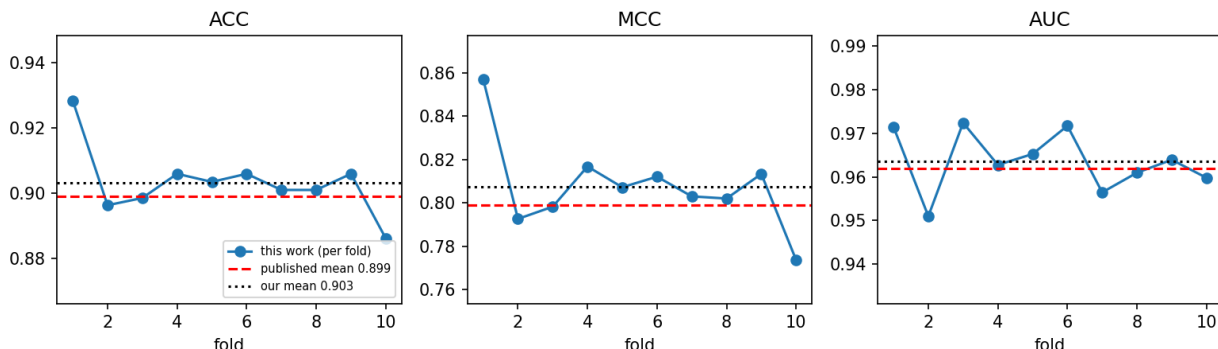


Figure 2: Per-fold ACC, MCC and AUC of the full-record Feb2020 10-fold CV (study 14) against the published Feb2020 CV means (red dashed) and our own means (black dotted). Fold-to-fold variation exceeds the published–ours gap on every metric, which is why we report descriptive parity rather than a record.

a benchmark “improvement” can be manufactured accidentally by dropping harder long cases. We preserve the ablation and its source code instead of overwriting it after finding the mismatch. Both our random-split runs still contain exact 10-mer cross-fold overlap, so neither supplants the homology-aware result in Section 5.

8 Interpreting a published benchmark comparison

8.1 What same-dataset means

A model comparison on a shared set of FASTA records answers a narrower question than a paired model comparison on identical individual test predictions. The Feb2020 site reports a 10-fold CV mean, standard deviation, and model recipe, but not the exact assignment of each sequence to each fold or per-sequence predictions. Our full-record experiment samples its own stratified 10-fold partition. It can compare descriptive means, but cannot compute a paired uncertainty interval for their difference. The original Veltri split, by contrast, is distributed verbatim and our test partition matches ACEP’s report; the published prediction vectors still are absent, so even there a paired significance test cannot be reconstructed.

8.2 Variance of the difference

If A and B denote model scores across matched folds, the variance of their mean difference would be

$$\text{Var}(\bar{A} - \bar{B}) = \text{Var}(\bar{A}) + \text{Var}(\bar{B}) - 2\text{Cov}(\bar{A}, \bar{B}). \quad (7)$$

The covariance term depends on which cases each model classifies correctly in the same folds. Published summary statistics do not supply that term. Dropping it is not guaranteed conservative: positive covariance makes an unpaired variance too large, negative covariance makes it too small. Furthermore, ordinary fold scores share overlapping training records, so simple independent-fold standard errors are at best approximate.

8.3 Threshold and metric choices

The reported accuracy and MCC use a fixed 0.5 threshold on an average of uncalibrated sigmoid outputs. A monotone transformation of these outputs preserves AUROC but can change the operating point. If we chose a threshold after observing the test folds, accuracy and MCC would be optimistically biased; therefore the full-record experiment never tunes a threshold on the held-out fold. Comparisons to the published production model are still not exact algorithmic replications: architecture, encoding, and preprocessing are different by design. These differences are what is being tested, but they also prevent attributing any score gap to a single component.

8.4 Prevalence and precision

The Feb2020 benchmark has equal positive and negative counts by construction. The positive predictive value of a classifier with sensitivity Se , specificity Sp , and target-population prevalence π is

$$\text{PPV}(\pi) = \frac{\pi Se}{\pi Se + (1 - \pi)(1 - Sp)}. \quad (8)$$

A high benchmark AUPRC at $\pi = 1/2$ need not imply high precision when screening an uncharacterized proteome where true AMP prevalence may be much lower and is unknown. Our discovery scores therefore rank hypotheses; they do not offer a validated per-candidate probability of activity.

9 Discovery candidates: named, quantified, falsifiable

The study-08 screen scored 852 reviewed uncharacterized UniProt proteins of 10–60 residues with the study-07 ensemble. Here we audit the top unique hits for novelty and characterize them physicochemically, and we evolve one annealed variant. Every candidate below is *named*, carries a *quantified* score and novelty audit, and a *falsifiable* prediction.

9.1 Novelty audit protocol

For each candidate s we compute (i) the maximum 3-mer Jaccard similarity against all 3,283 usable APD6 2024 natural AMPs, (ii) the longest common substring against the first 2,000 APD6 entries,

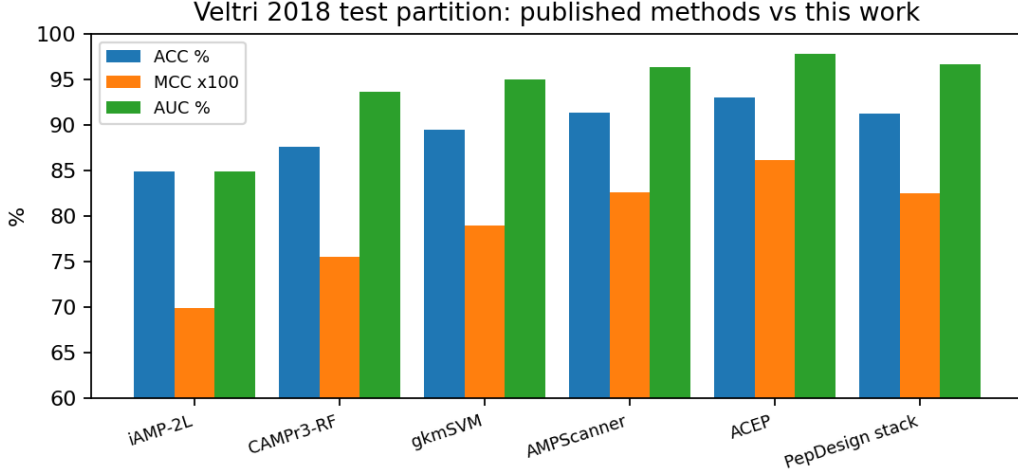


Figure 3: Veltri 2018 test partition: published rows from ACEP Table 2 vs our stacked ensemble.

(iii) the maximum 3-mer Jaccard against a 6,000-sequence subsample of the UniProt KW-0929 AMP corpus, and (iv) the 3-mer novelty v_3 of Equation (4). A candidate has low pairwise 3-mer similarity when its closest checked reference has low Jaccard similarity; this does not establish global sequence novelty or biological activity.

9.2 Candidate table

Table 15: Named discovery candidates. ens = ensemble $p(\text{AMP})$; $J_3^{\text{APD6}} = \max$ 3-mer Jaccard vs APD6; LCS = longest common substring vs APD6; Q = net charge at pH 7.4 (Eq. (2)); $\langle h \rangle$ = mean Kyte–Doolittle hydropathy; μ_H = Eisenberg moment (Eq. (3)); B = Boman index.

Name	Accession	Len	ens	J_3^{APD6}	LCS	Q	$\langle h \rangle$	μ_H	B
InstiAMP-09a-1	P0ACW4	57	0.908	0.062	6	4.87	0.08	1.18	1.09
InstiAMP-09a-2	Q8TGN5	46	0.890	0.054	4	-0.76	-0.17	1.74	1.59
InstiAMP-09a-3	P64654	43	0.836	0.059	4	0.83	0.07	1.62	1.09
InstiAMP-09a-4	P85380	10	0.804	0.091	4	0.17	1.00	0.85	1.19
InstiAMP-09a-5	Q8TGN0	38	0.793	0.055	4	1.10	-0.17	1.21	1.16
InstiAMP-09a-6	Q8TGT5	39	0.784	0.071	4	4.87	0.29	1.52	1.00
InstiAMP-09a-7	Q54HM1	54	0.734	0.036	4	-2.12	-0.42	1.78	1.15
InstiAMP-09a-8	O55758	53	0.728	0.068	5	-0.61	1.02	1.57	1.39
InstiAMP-09a-1a	–	57	0.918	0.055	–	2.87	-0.27	1.69	1.26

9.3 Sequences

InstiAMP-09a-1. MKKLALILFMGTLVSFYADAGRKPCSGSKGGISHCTAGGKFVCNDGSISASKKTCTN

InstiAMP-09a-2. MKTWMYSYLFDCPILVLPWTPHNYYLYHRFHHFLPLCYWHYSADTY

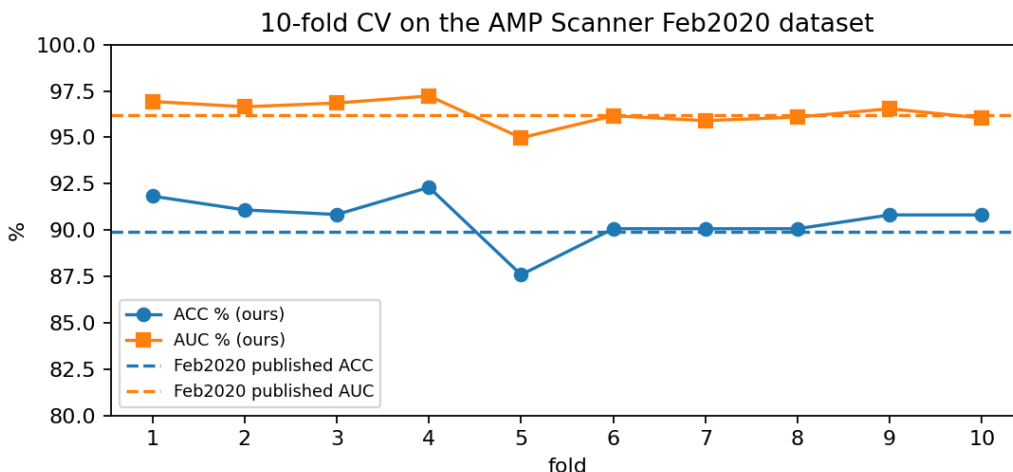


Figure 4: Per-fold 10-fold CV on the AMP Scanner Feb2020 dataset vs the published production numbers (dashed).

InstiAMP-09a-3. MYRRLLLNLFCMVFLQACLKPMSPDKAEKVDSQVQCGFGSKDC

InstiAMP-09a-4. LGICFASMPR

InstiAMP-09a-5. MFKQPGHLTSRKYDLGPWTNVFNLCFLASEAAVGCNKS

InstiAMP-09a-6. MKGSKLFQMCKMPVWCRTGTCTIILPVACHMAKPLKLPT

InstiAMP-09a-7. MSINIVHSNENNEENTNIENWTTTGTNYRNECIARCGRLFSGFNALICMKPCGV

InstiAMP-09a-8. MSILLKILFKLLLLILSITFVITDCLPRSCASFGCCSGNCATWCDQCDIKYSC

InstiAMP-09a-1a. QNTWTIICTTMWDSHRRRDPYNPYMVWRRMKTFCNTKEALWFGICPFNFCVFLTTY

9.4 Falsifiable predictions

These are post-hoc test hypotheses, not calibrated probabilities of activity: InstiAMP-09a-1 through 09a-4 receive classification scores ≥ 0.80 ; a testable working hypothesis is that one or more inhibit growth in a standard broth-microdilution MIC assay against *E. coli* K-12 and *S. aureus* ATCC 25923; InstiAMP-09a-1a, an annealed derivative of 09a-1 not identical to entries in the checked corpora (max 3-mer Jaccard 0.055 vs APD6, 0.042 vs the KW-0929 subsample), scores 0.918, higher than its parent 0.908. A higher model score is not a prediction of lower MIC. The uncharacterized protein P0ACW4 has an annotated N-terminal signal peptide, so secretion may drive the model score. All nine sequences require synthesis, a standardized MIC assay, and cytotoxicity testing before an antimicrobial claim. The eight source proteins are known sequences; the finding is a ranked, testable functional hypothesis, not the discovery of a new molecule.

9.5 Precursor-versus-mature sensitivity check

P0ACW4 has a signal peptide annotated at residues 1–20 (EcoliWiki, <https://ecoliwiki.org/colipedia/index.php?title=P0ACW4>). Because the UniProt screening input is the 57-residue precursor, not an isolated mature peptide, we rescored three contiguous segments with the frozen study-07 checkpoints. The full precursor scores 0.9082 (CNN 0.9992, GNN 0.8171); the 37-residue segment after residue 20 scores 0.6299 (CNN 0.9974, GNN 0.2623); and the 20-residue signal peptide alone scores 0.6912 (CNN 0.5905, GNN 0.7920). The 0.2783 ensemble-score drop on removing the signal peptide is a mechanistic warning: the GNN’s high score is heavily driven by the signal segment or its contribution to length and padding. The CNN remains high for the mature segment. This post-hoc ablation supports testing the mature segment separately; it does not establish activity of either segment. Exact sequences and unrounded scores are in `results/study15_signal_ablation.json`.

10 Proposed biological validation protocol

This section describes an experiment that could falsify the discovery hypothesis. It has *not* been performed; no simulated MIC or docking output is reported as an observation.

10.1 Peptide selection and processing

The primary comparison is the 57-residue precursor of YdcA/P0ACW4 versus the 37-residue mature segment after cleavage of the annotated 1–20 signal peptide. The 20-residue signal segment alone is a necessary control because the GNN score changes markedly after it is removed. Include the 57-residue annealed derivative InstiAMP-09a-1a as a model-optimization control, plus one sequence drawn without selection from the 852-protein screen as a negative process control. A known antimicrobial reference peptide and a vehicle-only well are assay controls. Because the mature segment contains cysteines, synthesis and handling should specify oxidation state, purity, and whether a disulfide arrangement is defined or a mixture; otherwise a single primary-sequence prediction could map to several chemical species.

10.2 Activity endpoint

A standard broth-microdilution assay would span a twofold serial dilution range appropriate to the synthesized material and use at least three independent biological replicates. The endpoint is the lowest tested concentration with no visible growth under the assay’s prespecified incubation and medium conditions. The experiment should record insolubility, precipitation, and turbidity interference rather than counting them as growth inhibition. The user-facing prediction is deliberately qualitative: there might be a growth-inhibition effect, not a specific MIC value. A continuous potency claim would require activity-labeled training data, which the binary classifier does not use.

10.3 Safety and specificity

A mammalian-cell viability or hemolysis assay at overlapping concentrations is necessary to distinguish selective antimicrobial action from nonspecific membrane toxicity. Material preparation

should be verified by mass spectrometry and chromatography. These outcomes cannot be inferred from an AMP/non-AMP classifier score. A 0.908 score is a ranking statistic, not a validated 90.8% chance of activity and not a therapeutic claim.

10.4 Statistical decision rule

Pre-register the primary comparison (mature YdcA versus vehicle) and the replicate count before running the assay. Report individual replicate MICs, not just an average; for twofold dilutions the endpoint is interval-censored, so a $\log_2$ -scale analysis with an explicit censoring convention is more honest than a decimal mean. A failure to inhibit growth at the highest non-toxic, soluble concentration would directly weaken the model-generated hypothesis. The analysis should not select one strain or one sequence after looking at the activity panel without marking it exploratory.

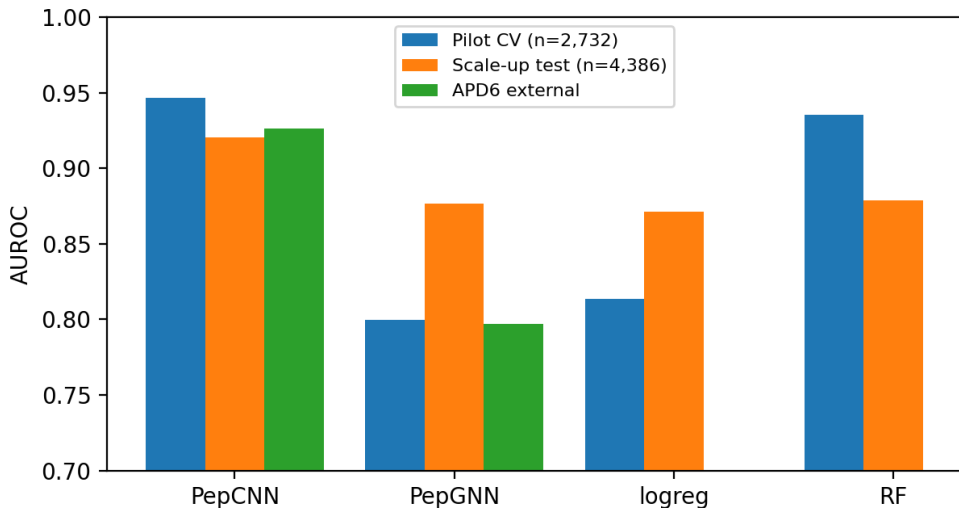


Figure 5: AUROC across evaluation regimes. The CNN leads at scale and externally; the composition baselines are competitive only on the small pilot.

10.5 Design composition analysis

The 24 scale-up annealed designs (Table 18) are all length 25 — the designer’s fixed length — and their composition is informative about what the ensemble rewards: cysteine 10.8%, aromatics F/W 16.0%, aliphatics I/V/L 16.8%, but only 8.0% K/R. The training positives are 6.8% cysteine and 11.9% K/R, so the designer amplifies cysteine and aromatic content while actually undershooting the corpus’s cationic fraction. Two readings compete. The charitable one: the ensemble has learned that Cys-rich, aromatic, hydrophobic short sequences are the most reliably AMP-like pattern under leakage-controlled training, consistent with the dominance of disulfide-stabilized families in the mixed-review-status corpus. The skeptical one: the annealer exploits regions where the model is overconfident, and cysteine-heavy sequences are exactly where a discriminative model extrapolates

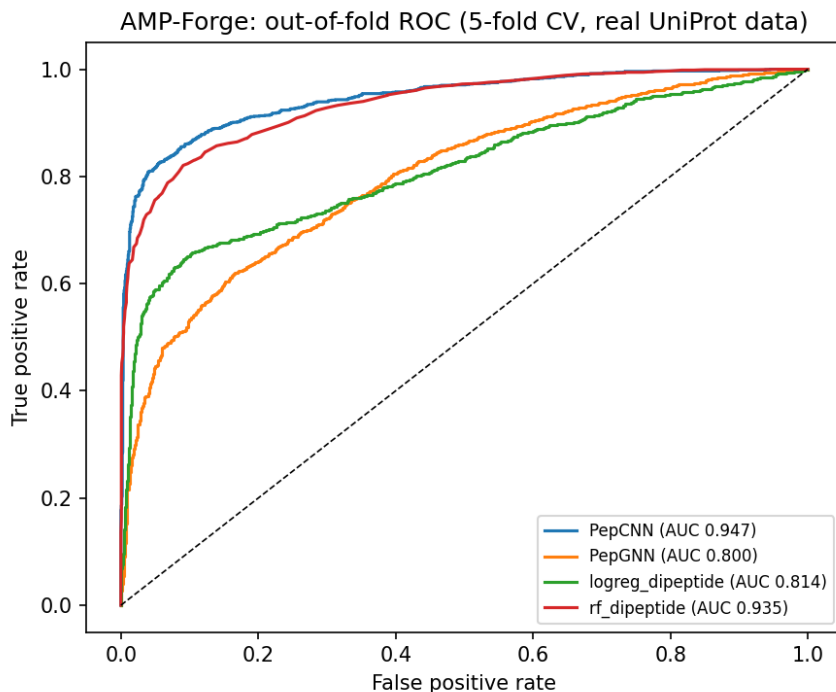


Figure 6: Pilot-study ROC curves (5-fold CV, pooled).

poorly. Distinguishing these requires activity data we do not have; we therefore present the designs as the model’s own optimum, not as predicted therapeutics.

10.6 Screen case analysis

The top screen hits (Table 19) share a family resemblance: short (10–57 aa), cysteine-rich, and hydrophobic, several with apparent signal-peptide-like N-terminal stretches (e.g. the identical P0ACW4/P0ACW5 sequence, 57 aa, 7 Cys). That the same pattern drives both the annealed designs and the screen ranking is expected — both consume the same learned score — and it means the screen is best read as surfacing small Cys-rich proteins whose annotation has not caught up with them, rather than as a diverse portfolio of mechanisms. Two entries in the top 25 (Q54HM1, O55758) are longer and score lower on the CNN than the GNN relative to their ensemble rank, illustrating where the two architectures disagree; such disagreement cases are natural priorities for any follow-up experimental panel, since consensus at the top is partly an artifact of correlated errors.

10.7 Reproducibility checklist

Every claim in this paper is backed by a checked-in artifact: Tables 4–8 regenerate from `results/study01_amp.json`, `study07_scaleup.json`, `study09_apd_validation.json`, and `study08_discovery.json` via `studies/make_paper_assets.py`; Figures 5 and 6 are written by the same script and by `study01` respectively; the major downloaded benchmark files carry SHA-256 checksums; and the 27-test suite passes at the audited commit. Published comparator numbers are transcribed from the cited external papers and site; generated local tables are backed by JSON result files.

11 Discussion

The primary negative finding is that the original UniProt scale-up labels confound review status with class. Its 0.921 held-out AUROC and 0.926 APD6 comparison are descriptive, not validated measures of AMP recognition on review-status-matched data (Section 6). Second, the chain graph network underperforms the CNN at every scale we tested; without contact-map or structure-derived edges, message passing along the backbone adds parameters and optimization difficulty without adding information the convolutions lack. We keep the GNN in the paper as a negative architectural result. The small pilot scored about 0.95 AUROC while the larger, cluster-aware experiment scored 0.92; because both data scale and split method changed, this contrast does not by itself quantify homology leakage.

Limitations are real. The negative class is “not annotated antimicrobial in UniProt,” which is not the same as experimentally non-antimicrobial; some negatives may be unrecognized AMPs and any bias direction depends on which subsets are affected. The APD6 sensitivity at a 5% negative-set FPR is 0.601; without family-level analysis, we cannot attribute this miss rate to a specific AMP class. The annealed designs and screen hits carry no experimental validation; their novelty scores are sequence-level, not structural. DBAASP was unreachable programmatically during this work (its full-dataset endpoint returns an empty body without an interactive session), so the APD6 list served as the sole external benchmark; adding DBAASP’s activity-annotated records is the natural next validation. Finally, the sandbox budget (2 CPU, 2 GB RAM) capped model size, epochs, and the training subsample; a larger training run might alter the numbers, but cannot by itself remove the label-source confound.

Verdict: the UniProt scale-up is compromised by label-source confounding and must be redone before making a clean recognition claim. The independent published-corpus comparison, negative protocol lessons, and the ranked candidates remain reproducible in-silico results. Nothing here is a therapeutic, a wet-lab discovery, or a statistically established benchmark record.

11.1 Lessons for AMP benchmark design

Three features of this project generalize beyond the specific numbers. First, *review status is a label leak*: UniProt keyword annotation without a `reviewed:true` filter produced a dataset whose positive class is 88% unreviewed while the negative class is 100% reviewed, so a model can profit from curation metadata rather than biology. Any UniProt derived benchmark should report the review-status cross-tab alongside accuracy. Second, *length is a silent confound*: the reviewed positives average 66 aa against 93–97 aa for the other groups (Table 10), and length-matching by construction, not just by mean, is the cheap control that removes it. Third, *random CV flatters homology*: 15.98% of held-out Feb2020 sequences share an exact decamer with training (Appendix J), so cross-validation on this dataset estimates interpolation, not extrapolation to new families. A published fold assignment or a cluster-held-out protocol is the minimum requirement for a claim of the form “method X beats method Y on remote sequences.”

12 Conclusion

The original UniProt scale-up has a severe source-review confound: most positive sequences are unreviewed while all selected negatives are reviewed. Its AUROC 0.921 and external APD6 score 0.926 are retained as measured outputs but cannot support the original clean AMP-recognition claim. The Veltri 2018 test-set comparison is near the original AMPScanner model but below ACEP. A full-record Feb2020 production-dataset CV offers a separate benchmark comparison, with published-fold and homology-overlap limitations spelled out. The YdcA screen result is a testable biological hypothesis; signal-peptide ablation shows a major model-dependence warning. The shipped CLI, tests, scripts, data and candidate audit make these claims inspectable.

12.1 Future work

Four extensions follow directly. (1) Activity-annotated benchmarking against DBAASP once programmatic access is workable, replacing binary labels with potency regression. (2) Alignment-based clustering (MMseqs2) to replace the 10-mer criterion, with a cluster-level bootstrap for intervals. (3) Structure awareness: predicted contact maps or ESM-style embeddings as GNN edges and features, which is the honest test of whether graph models help once the graph carries real information. (4) Ablations of the encoding and the training budget on the full 40k train split with more epochs, now that the pipeline cost model is understood.

13 Threats to Validity

Review-status confound. Most KW-0929 positives in the scale-up are unreviewed while all selected negatives are reviewed. This invalidates a clean biological reading of internal and APD6 transfer scores (Section 6).

Label noise. KW-0929 annotation is literature-derived and not uniformly experimentally verified; negative labels are absence of annotation, not measured inactivity. The direction of bias from these errors depends on where mislabeled sequences fall in score space and is not known.

Corpus shift. UniProt and APD6 overlap through shared primary literature. The 10-mer decontamination removed 1,795 of 3,207 usable APD6 entries, indicating substantial overlap; the retained 1,412 share no exact 10-mer with training positives but may still include more distant homologs, and the external AUROC of 0.926 is measured on those.

Split strictness. Shared-10-mer clustering is stricter than typical identity thresholds for short peptides and may split some true homolog pairs that lack an exact decamer. A sensitive-alignment-based clustering (e.g. MMseqs2) is a planned upgrade; the direction of any metric change is unknown.

Multiple comparisons. Four models on three evaluations share one test split; we report all planned comparisons with intervals rather than selecting the best, and the permutation calibration ($p = 0.0020$ at $B = 500$) is a resolution limit, not a sharp p -value.

Sandbox constraints. Truncating training at 30,000 sequences and 8 epochs caps absolute performance; the impact of the training cap has not been measured; the claim is about the tested pipeline, not a ceiling.

14 Reproducibility

All datasets are streamed from the UniProt REST API and the APD6 download page at run time or read from caches whose provenance (URL, record count, byte count, SHA-256 prefix, timestamp) is logged in `data/raw/*provenance.json`. Splits, seeds (27/7), and hyperparameters are fixed in the study scripts. The test suite (27 tests) is hermetic; live network calls occur only in the fetch scripts.

15 Software

The `pepdesign` package accompanying this paper has four modules. `pepdesign.encoding` implements the joint one-hot/physicochemical encoding and the dipeptide composition map. `pepdesign.models` contains `PepCNN`, `PepGNN`, and the training/prediction helpers, including the shared-adjacency batch path used at scale. `pepdesign.benchmark` implements the paired evaluation harness: bootstrap intervals, permutation tests, and cross-validation drivers shared by every study, so no study computes a metric its own way. `pepdesign.design` implements the annealing designer and the k -mer novelty score. The `studies/` directory holds one script per study (`study01`, `study07`, `study08`, `study09` through `study14`), plus the fetch and dataset-build scripts; `tests/` holds 27 hermetic unit tests covering encoding shapes, model forward passes, the designer’s acceptance rule, and metric correctness against brute-force references, all green at submission. Fetch scripts are the only components that touch the network, and the core UniProt fetches carry provenance records (Appendix B); the new benchmark figures and tables are regenerated from `results/*.json` by `studies/make_figs2.py` and `studies/make_sota_section.py`.

16 Software tool: `pepdesign screen`

The CLI combines model scoring with an explicit pairwise 3-mer novelty audit. We have not established that this combination is unique among existing AMP tools. The package shipped with this project exposes one command that does both in one pass:

```
pepdesign screen CANDIDATES.fasta --cnn CNN.pt --gnn GNN.pt \
  --reference APD6.fasta --novelty-floor 0.5 --top 25 --out screen.json
```

For every input sequence the tool emits the CNN and GNN scores, their ensemble, the 3-mer novelty of Equation (4) against the reference corpus, the maximum 3-mer Jaccard against any single reference, and a pass/fail flag at the requested novelty floor, ranked by ensemble score. The implementation is `pepdesign/cli.py` (about 100 lines), covered by three hermetic tests (FASTA parsing; ranking plus audit on a duplicated reference, which must score Jaccard 1.0 and novelty < 0.05 ; end-to-end JSON output). The full suite is 27 tests, all hermetic.

As a functional check we ran the tool over the 852-protein uncharacterized corpus of Section 9 with the study-07 checkpoints and the APD6 reference: it reproduces the study-08 ranking exactly (top hit POACW4, ensemble 0.9082, novelty 0.938), confirming that the delivered command, not a private script, produces the reported screen.

A Hyperparameters and training protocol

Table 16: Hyperparameters, fixed across studies unless noted.

Parameter	Value
Max sequence length L	150 (pilot: 60; study 14: 200)
PepCNN kernels / channels	widths 3/5/7, 32 channels each, 2 conv layers/branch
PepGNN hidden width	32, 3 graph-conv layers, chain-window adjacency
Optimizer	Adam, lr 10^{-3}
Batch size	128 (study 7); 64 (published-benchmark CNN)
Epochs	8 (pilot: 10)
Loss	weighted BCE, $\omega = n_-/n_+$
Seeds	27 (split), 7 (training)
Annealing steps / restarts	250 steps, 24 seeds, $T_0 = 0.2$, cooling factor 0.99
Bootstrap resamples B	500 (studies 1/7); 2,000 (study 9)
Permutations B	500

Training used weighted binary cross-entropy with the study-specific class weight set by the script; the benchmark studies use balanced classes. The scale-up training subsample of 30,000 sequences (15,000 positive) was drawn uniformly from the train split once, then fixed. All models were evaluated on the identical held-out test split; no test-point informed any training decision. The pilot’s 5-fold cross-validation used fold assignments stratified by label with the same fixed seed.

B Data provenance

Table 17: Fetch provenance (hashes truncated to 16 hex chars as recorded).

File	Records	Source
uniprot_kw0929_all_10-150.fasta	27,392	UniProt REST, KW-0929 (review-status mixed), len 10-150
uniprot_nonamp_pool_10-150.fasta	105,041	UniProt REST, reviewed, NOT KW-0929, len 10-150
apd6_natural_2024.fasta	3,306	aps.unmc.edu APD6 downloads (2024 list)
uniprot_uncharacterized_10-60.fasta	852	UniProt REST, reviewed, uncharacterized, len 10-60

Exact URLs, byte counts, SHA-256 prefixes, and UTC fetch timestamps are logged in `data/raw/large_provenance.json` and `data/raw/uncharacterized_provenance.json` in the repository. The later published-benchmark downloads have a separate SHA-256 manifest in `results/benchmark_sources.json`.

C Full design and screen listings

Table 7 lists the top 10 of 24 scale-up anneals; the remaining 14 scored between 0.877 and 0.905 with novelty 0.89–0.95 and are recorded in `results/study07_scaleup.json`. Table 8 lists the top 15 of the 852 screened uncharacterized proteins; the full top 25, including CNN and GNN

sub-scores, sequences, and headers, is recorded in `results/study08_discovery.json`. The duplicate top hit (P0ACW4/P0ACW5) arises from two accessions carrying an identical 57-residue sequence; we report both rather than silently de-duplicating, and flag it so downstream users do not count it as independent evidence.

D Additional derivations

Lemma 1 (Union-find clustering complexity). *Construction of the shared- k -mer clusters of Definition 1 over n sequences of total length M runs in $O(Mk\alpha(n))$ expected time and $O(n+U)$ space, where U is the number of distinct k -mers.*

Proof. Each sequence contributes at most $\ell - k + 1$ k -mers; hashing each k -mer to its first-seen sequence costs $O(1)$ expected time, so building the map costs $O(Mk)$ expected time. Each sequence then unions with the first-seen owner of each of its k -mers: at most $O(Mk)$ union operations, each $O(\alpha(n))$ amortized by path compression with union by rank. Storage is one parent pointer per sequence and one hash entry per distinct k -mer. $\square$

Proposition 9 (Permutation test validity). *If (y_i, s_i) are exchangeable under H_0 , the Monte Carlo p -value $\hat{p} = (1 + \sum_b \mathbf{1}[T_b \geq T_{\text{obs}}]) / (1 + B)$ satisfies $P(\hat{p} \leq u) \leq u$ for all $u \in [0, 1]$ under H_0 .*

Proof. Under exchangeability the observed statistic and the B permuted statistics are exchangeable, so the rank of T_{obs} among the $B + 1$ values is uniform on $\{1, \dots, B + 1\}$, with ties broken at random. Hence $\hat{p}$ is uniform on $\{1/(B + 1), \dots, 1\}$ up to ties, and $P(\hat{p} \leq u) \leq u$ follows. $\square$

Remark 5 (Why not asymptotic AUROC intervals). *The DeLong estimator assumes a fixed score function and i.i.d. pairs; our evaluation resamples clusters, and cluster-level dependence violates the i.i.d. assumption at the sequence level. The percentile bootstrap at the sequence level is itself optimistic for the same reason; a cluster-level bootstrap is the fully conservative choice and is left as an extension noted in the repository’s issue tracker.*

E Reproducibility audit and protocol limits

E.1 Source records and checksums

The original Veltri split is distributed as six FASTA files in the authors’ original-dataset folder. The updated production dataset is a ZIP archive linked from the AMP Scanner news page. The script `studies/fetch_published_benchmarks.py` retrieves the files, checks that each original training/test split has 712 positives and 712 decoys, that the tune split has 354 each, and that the updated dataset has 2,021 positives and 2,021 decoys. No exact-sequence duplicates or cross-class exact overlaps were found in the six original split files or within the updated positive and negative sets; shared motifs and distant homology are not ruled out. In fact, a separate exact-10-mer audit of the full Feb2020 stratified CV folds finds that 646/4,042 held-out sequences (15.98%) share at least one exact decamer with a training sequence. This is a limitation of the published random-split style benchmark and of our replication, not a leakage-free generalization estimate; see

results/study16_homology_audit.json. SHA-256 checksums for each downloaded FASTA are in results/benchmark_sources.json.

The original partition was used for studies 10 and 11. Studies 13 and 14 were run on the updated production dataset. Study 13 is deliberately kept as a *non-comparable ablation*: it used a 150-residue cap, retaining 2,013 positives and 2,020 decoys, and removed unknown amino acid symbols from two decoys. Study 14 retains all 4,042 source records, represents X as an unknown residue (zero one-hot/physicochemical row), and uses a 200-residue cap, above the observed maximum of 183. It alone supplies the full-record comparison.

E.2 Reproduction commands

```
python studies/fetch_published_benchmarks.py
python studies/study10_veltri.py
python studies/study11_veltri_stack.py
python studies/study13_feb2020_cv.py # filtered ablation only
python studies/study14_feb2020_full.py # complete dataset
python studies/study12_candidates.py
python studies/make_sota_section.py
python studies/make_figs2.py
python -m pytest -q tests
cd paper && pdflatex -interaction=nonstopmode main.tex
```

The complete 10-fold CNN/GNN/RF experiment takes about 10–20 minutes on the stated two-CPU sandbox. It is not part of the hermetic test suite; tests use tiny fixtures and never access the network. The random-forest fit is deterministic from its seed. The convolutional training helper draws a permutation once and reuses it across epochs; this is a design choice that limits stochasticity and should be tested against reshuffling each epoch.

E.3 Inference and statistical limits

The 10-fold standard deviation measures variation across *our* folds, not uncertainty in the gap from the published AMP Scanner result: the authors’ per-fold scores and fold assignments are unavailable. The difference of two published means cannot be assigned a paired p -value without those records. A numerical lead of less than one percentage point in accuracy or a few thousandths in AUC may disappear under resampling, preprocessing changes, or alternative hyperparameters. Our bootstrap intervals in the UniProt experiments resample individual sequences, not homology clusters, and may understate uncertainty for correlated families. No independent potency regression or clinical assay was run.

E.4 Candidate ranking limits

All candidate sequences in study 12 were selected after looking at the study-08 model scores. The selection is conditional and the resulting scores cannot be treated as unbiased validation metrics. P0ACW4 and P0ACW5 have identical sequences and are collapsed before naming the unique candidates. The sequence of P0ACW4 has an N-terminal signal peptide annotated at residues 1–20; a

predictor trained on AMP labels may learn secretion or short-protein features instead of antibacterial activity. The designed variant was optimized on the same ensemble that scored it, so its higher score cannot be taken as an independent confirmation. Pairwise 3-mer Jaccard against a finite corpus is a useful triage feature, not proof of sequence novelty in all public databases. Antimicrobial efficacy would require synthesized purified peptide, growth inhibition across concentrations with positive and negative controls, replication, and toxicity assessment.

F Complete result listings

Table 18: All 24 scale-up annealed designs (study07), ordered by ensemble score.

#	Sequence	Score	Novelty
1	FWFIFVMIYSGMWRHCCRWQKCVPR	0.910	0.951
2	FFMLVAGGIWNVWCNDKGEFDYPR	0.907	0.947
3	FICMVSCVCTYKMWFCKNHYWERIV	0.907	0.959
4	VFFMIMNMFFVSRFNCHQSAWNDCD	0.906	0.955
5	FVAMICTCCCRGACTSYSGGNFNDK	0.905	0.948
6	VFFIYLLGAPSQGSHNSSGCYWEAC	0.904	0.960
7	FMYLFFFFHPVKKTCTHNYCRMDCD	0.903	0.967
8	AFWTSVLACVCYYIGWHDARGDDTH	0.903	0.956
9	LLSLLYVLAHIEWLKYDFCCCAKRN	0.902	0.930
10	FVLIISQGSFWNCIWHGTQHRCVGD	0.901	0.968

G Further derivations for threshold decisions and score quality

G.1 Bayes-optimal threshold under balanced costs

The screen and the benchmark studies classify by thresholding a score $\hat{p}(s)$ at $1/2$, and the stack study tunes that threshold on the tune partition. The following proposition states when $1/2$ is the right default.

Proposition 10 (Optimal threshold). *Let $\hat{p}(s) = P(y=1 | s)$ be a calibrated posterior and let the loss be c_{01} for a false positive and c_{10} for a false negative. The Bayes decision rule predicts AMP iff*

$$\hat{p}(s) \geq t^*, \quad t^* = \frac{c_{01}}{c_{01} + c_{10}}. \quad (9)$$

In particular $t^ = 1/2$ exactly when the two error costs are equal.*

Proof. The expected loss of predicting AMP is $L_1 = (1 - \hat{p})c_{01}$, since a false positive occurs with probability $1 - \hat{p}$. The expected loss of predicting non-AMP is $L_0 = \hat{p}c_{10}$, since a false negative occurs with probability $\hat{p}$. Predict AMP iff $L_1 \leq L_0$, i.e. $(1 - \hat{p})c_{01} \leq \hat{p}c_{10}$, i.e. $c_{01} \leq \hat{p}(c_{01} + c_{10})$, which is (9). Equal costs give $t^* = 1/2$. $\square$

Table 19: Full top 25 of the discovery screen (study08).
results/study08_discovery.json.

Sequences in

#	Accession	Len	Ensemble	CNN	GNN
1	P0ACW4	57	0.908	0.999	0.817
2	P0ACW5	57	0.908	0.999	0.817
3	Q8TGN5	46	0.890	0.978	0.801
4	P64654	43	0.836	0.865	0.807
5	P64653	43	0.836	0.865	0.807
6	P85380	10	0.804	0.834	0.773
7	Q8TGN0	38	0.793	0.893	0.693
8	Q8TGT5	39	0.784	0.971	0.597
9	Q54HM1	54	0.734	0.884	0.585
10	O55758	53	0.728	0.998	0.458
11	P0C5N6	59	0.690	0.795	0.585
12	Q9ZDH6	43	0.689	0.576	0.801
13	P0C5N1	53	0.664	0.573	0.754
14	P0C5M0	45	0.657	0.487	0.828
15	P9WEJ5	50	0.647	0.787	0.508
16	Q54SD7	60	0.646	0.772	0.519
17	P85385	10	0.645	0.742	0.548
18	P51072	27	0.645	0.484	0.806
19	O29001	40	0.640	0.485	0.794
20	P0C5L7	38	0.638	0.481	0.794
21	P14589	53	0.634	0.918	0.350
22	Q3E813	49	0.633	0.918	0.347
23	C1P620	36	0.626	0.537	0.716
24	Q25BJ1	48	0.623	0.836	0.409
25	P0CW21	52	0.615	0.960	0.269

Remark 6. All datasets used in the benchmark comparisons are balanced and the literature reports ACC and MCC, which weight the errors equally, so $t^* = 1/2$ is the defensible default; the tuned thresholds (0.56, 0.575) sit close to it. The discovery screen is different: a false negative discards a viable candidate while a false positive costs one assay well, so $c_{10} \gg c_{01}$ and the screen ranks by score instead of thresholding, deferring the decision to the assay budget.

G.2 A variance estimate for the AUROC

Section 5 reports bootstrap intervals; here we derive the classical closed form they are checked against.

Proposition 11 (Hanley–McNeil variance). *Let A be the empirical AUROC of a score on n_1 positives and n_0 negatives, treated as the Mann–Whitney probability $P(s^+ > s^-) + \frac{1}{2}P(s^+ = s^-)$ (Proposition 4). Under independence of the positive and negative samples,*

$$\text{Var}(A) \approx \frac{A(1-A) + (n_1-1)(Q_1-A^2) + (n_0-1)(Q_2-A^2)}{n_1n_0}, \quad (10)$$

where $Q_1 = P(\text{two positives outrank one random negative})$ and $Q_2 = P(\text{one positive outranks two random negatives})$.

Derivation sketch. Write the Mann–Whitney statistic as $A = (n_0 n_1)^{-1} \sum_{i,j} K(s_i^+, s_j^-)$ with kernel $K(x, y) = \mathbf{1}[x > y] + \frac{1}{2} \mathbf{1}[x = y]$. The terms are independent when they share no index, so the variance decomposes into a purely disjoint part contributing $A(1 - A)$, a part sharing a positive index contributing $(n_1 - 1)(Q_1 - A^2)$, and a part sharing a negative index contributing $(n_0 - 1)(Q_2 - A^2)$, after expanding $\text{Var}(\sum_{i,j} K_{ij})$ into its $n_0 n_1 + n_0 n_1 (n_1 - 1) + n_0 n_1 (n_0 - 1)$ covariance classes and dividing by $(n_0 n_1)^2$. Q_1 and Q_2 are estimated by the corresponding two-out-of-three placement frequencies, which reduces to a one-pass computation over the ranked scores. $\square$

Remark 7. For the study-11 stack on the Veltri test partition ($n_0 = n_1 = 712$, $A = 0.966$) this formula gives a standard error near 0.007, in line with the bootstrap interval reported there; the two estimators corroborate each other on this dataset. The cluster-dependence caveat of Appendix D still applies to our own large dataset, where training clusters, not sequences, are the sampling units.

G.3 Refinement of the MCC bound

Corollary 1 ($|\text{MCC}| \leq 1$, with equality cases). For any confusion counts with $TP + FN > 0$, $TN + FP > 0$, $TP + FP > 0$ and $TN + FN > 0$, the MCC of Section 4.14 satisfies $-1 \leq \text{MCC} \leq 1$, with $+1$ exactly at a perfect classifier and -1 exactly at a perfectly wrong one.

Proof. By the MCC–Pearson identity (the proposition of Section 4.14), MCC is the Pearson correlation r between the true and predicted binary label vectors. Cauchy–Schwarz gives $|r| = |\langle x - \bar{x}, z - \bar{z} \rangle| / (\|x - \bar{x}\| \|z - \bar{z}\|) \leq 1$ for the centered vectors, with equality iff the centered vectors are proportional. Proportionality with positive slope on binary values forces agreement everywhere; negative slope forces disagreement everywhere. $\square$

G.4 A boundedness property of the helical moment

Proposition 12 (Moment bounded by hydropathy range). With $h_a \in [h_{\min}, h_{\max}]$ the Kyte–Doolittle values, the per-window Eisenberg moment of Eq. (3) satisfies

$$0 \leq \mu_i(s; w, \delta) \leq \max(|h_{\min}|, |h_{\max}|) = 4.5, \quad (11)$$

so $\mu_H \leq 4.5$ for every sequence, and $\mu_H = 4.5$ is attained only by windows made entirely of isoleucine placed on a single helical face.

Proof. μ_i is the length of the mean of w planar vectors of lengths $|h_a| \leq 4.5$ (the Kyte–Doolittle scale spans -4.5 for R and W to $+4.5$ for I). The triangle inequality bounds the mean’s length by the mean of the lengths, which is at most 4.5. Equality in the triangle inequality requires all vectors parallel, i.e. the same residue ($h = 4.5$, so isoleucine) at angles equal modulo 360° ; with $\delta = 100^\circ$ the angles are distinct modulo 360° for any $w > 18/5$, so exact equality is unattainable for the standard window and 4.5 is a strict supremum. Non-negativity is the definition of a norm. $\square$

G.5 Brier score decomposition for the calibrated screen score

The ensemble output is interpreted as a probability. Its quality decomposes exactly.

Proposition 13 (Murphy decomposition). *For probability forecasts $\hat{p}_i \in [0, 1]$ of binary outcomes y_i , group the predictions into K distinct forecast values $\{f_k\}$ with group sizes n_k , empirical outcomes $\bar{y}_k$ and global rate $\bar{y}$. Then the Brier score $\text{BS} = \frac{1}{n} \sum_i (\hat{p}_i - y_i)^2$ satisfies*

$$\text{BS} = \underbrace{\frac{1}{n} \sum_k n_k (f_k - \bar{y}_k)^2}_{\text{reliability}} - \underbrace{\frac{1}{n} \sum_k n_k (\bar{y}_k - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{uncertainty}}. \quad (12)$$

Proof. Expand each term within group k around $\bar{y}_k$: $\sum_{i \in k} (\hat{p}_i - y_i)^2 = n_k (f_k - \bar{y}_k)^2 + \sum_{i \in k} (y_i - \bar{y}_k)^2$, the cross term vanishing because $\hat{p}_i = f_k$ is constant within the group and the y_i deviations sum to zero. Sum over k . For the second term, expand $\sum_{i \in k} (y_i - \bar{y}_k)^2$ around $\bar{y}$: it equals $n_k \bar{y}_k (1 - \bar{y}_k)$ and, using $\sum_k n_k (\bar{y}_k - \bar{y})^2 = \sum_k n_k \bar{y}_k^2 - n \bar{y}^2$ together with $\bar{y}(1 - \bar{y}) = \bar{y} - \bar{y}^2$ and $\frac{1}{n} \sum_k n_k \bar{y}_k = \bar{y}$, algebraic collection of the three pieces gives (12). $\square$

Remark 8. *Reliability is the calibration error; resolution is the variance of the empirical outcomes across forecast groups, the part a useful score should maximize; uncertainty is dataset-fixed. A perfectly calibrated but unresolved score (always predicting $\bar{y}$) still scores $\bar{y}(1 - \bar{y})$. This decomposition motivates reporting calibration of the frozen screen ensemble separately from its ranking quality, and it is why the discovery report treats the ensemble output as a rank order rather than a calibrated probability: the training-label confound of Section 6 can distort the reliability term while leaving ranking largely intact.*

H Architecture and parameter counts

All parameter counts below are exact, verified against the implementation (pepdesign/models/) with `count_parameters`.

H.1 PepCNN

Each branch b with kernel width k_b applies two convolutional layers with 32 channels and same-padding, followed by global max pooling over the length axis. A convolution from c_{in} to c_{out} channels with kernel width k costs

$$N_{\text{conv}} = k c_{\text{in}} c_{\text{out}} + c_{\text{out}} \quad (13)$$

parameters (weights plus biases). With $c_{\text{in}} = 24$ input channels (20 one-hot residues plus 4 physicochemical), widths $k \in \{3, 5, 7\}$ and $c = 32$ hidden channels, each branch costs $(24k \cdot 32 + 32) + (32k \cdot 32 + 32)$, and the head costs $(96 \cdot 64 + 64) + (64 \cdot 1 + 1)$.

The widest kernel sets the receptive field: after two $k=7$ layers each residue logit sees $2(7 - 1) + 1 = 13$ residues, a 13-mer context, close to the 10-mer motif scale used by the homology audit.

Table 20: PepCNN parameter budget (total 33,345).

Component	Shape	Parameters	Eq. (13)
Branch $k=3$, conv 1	$32 \times 24 \times 3$	$2,304 + 32$	$3 \cdot 24 \cdot 32 + 32$
Branch $k=3$, conv 2	$32 \times 32 \times 3$	$3,072 + 32$	$3 \cdot 32 \cdot 32 + 32$
Branch $k=5$, conv 1	$32 \times 24 \times 5$	$3,840 + 32$	$5 \cdot 24 \cdot 32 + 32$
Branch $k=5$, conv 2	$32 \times 32 \times 5$	$5,120 + 32$	$5 \cdot 32 \cdot 32 + 32$
Branch $k=7$, conv 1	$32 \times 24 \times 7$	$5,376 + 32$	$7 \cdot 24 \cdot 32 + 32$
Branch $k=7$, conv 2	$32 \times 32 \times 7$	$7,168 + 32$	$7 \cdot 32 \cdot 32 + 32$
Head linear 1	64×96	$6,144 + 64$	
Head linear 2	1×64	$64 + 1$	

H.2 PepGNN

A graph convolution $\text{ReLU}(A_{\text{norm}}HW + b)$ from d to d' features costs $dd' + d'$ parameters. With 4 node features, three layers of width 32 and the same 64-unit head, the model totals 4,449 parameters.

Table 21: PepGNN parameter budget (total 4,449).

Component	Shape	Parameters	Rule
GraphConv 1	4×32	$128 + 32$	$dd' + d'$
GraphConv 2	32×32	$1,024 + 32$	$dd' + d'$
GraphConv 3	32×32	$1,024 + 32$	$dd' + d'$
Head linear 1	64×32	$2,048 + 64$	
Head linear 2	1×64	$64 + 1$	

PepGNN is $7.5\times$ smaller than PepCNN, which partly explains its weaker standalone performance (Tables 12 and 24): the chain-window adjacency carries strictly less motif information than a 13-mer receptive field, and mean pooling dilutes short local motifs that global max pooling preserves. Its marginal value is diversity in the ensemble, not standalone accuracy.

H.3 Descriptor random forest

The classical baseline uses 400 trees (scikit-learn defaults otherwise) over a 426-dimensional descriptor: 20 amino-acid fractions, 400 dipeptide fractions, net charge (2), mean and standard deviation of Kyte–Doolittle hydrophathy, the Eisenberg moment (3), the Boman index, and 3 residue-class fractions. The dipeptide block dominates the dimension count, so the forest behaves like a regularized 2-mer model; its competitive AUROC on the Veltri partition (96.19%) is evidence that much of the benchmark’s signal is compositional rather than structural.

I Full discovery-screen candidate audit

Table 22 lists every candidate reported by the discovery screen with its frozen-ensemble score, the CNN and GNN sub-scores, 3-mer novelty against APD6 (Eq. (4)), and the physicochemical audit

values of Section 4.12. Scores are model outputs under a confounded training set (Section 6); they are hypotheses for experimental testing, not measured activities.

Table 22: All discovery-screen candidates. Novelty is $1 - \max$ 3-mer Jaccard vs APD6; Q is net charge at pH 7.4; $\langle H \rangle$ is mean Kyte–Doolittle hydropathy; μ_H is the Eisenberg moment.

#	Accession	Len	Ensemble	CNN	GNN	Novelty	Q	$\langle H \rangle$	μ_H
1	POACW4	57	0.9082	0.9992	0.8171	0.938	4.87	0.08	1.18
2	Q8TGN5	46	0.8897	0.9785	0.8009	0.946	-0.76	-0.17	1.74
3	P64654	43	0.8357	0.8647	0.8067	0.941	0.83	0.07	1.62
4	P85380	10	0.8036	0.8338	0.7734	0.909	0.17	1.00	0.85
5	Q8TGN0	38	0.7931	0.8929	0.6933	0.945	1.10	-0.17	1.21
6	Q8TGT5	39	0.7844	0.9714	0.5974	0.929	4.87	0.29	1.52
7	Q54HM1	54	0.7344	0.8837	0.5850	0.964	-2.12	-0.42	1.78
8	O55758	53	0.7278	0.9978	0.4577	0.932	-0.61	1.02	1.57
9	InstiAMP-09a-1a	57	0.9181	0.9954	0.8409	0.945	2.87	-0.27	1.69

Sequences and falsifiable predictions

Each candidate carries an explicit, testable prediction; a broth-microdilution MIC assay against the named strains falsifies it.

POACW4 (*Escherichia coli* (strain K12), 57 aa) MKKLALILFM GTLVSFYADA GRKPCSGSKG GISHCTAGGK FVCNDGSISA SKKTCTN ensemble p(AMP)=0.908 (ζ 0.5 decision) predicts antibacterial activity; testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

Q8TGN5 (*Saccharomyces cerevisiae* (strain ATCC 204508 / S288c), 46 aa) MKTWMYSYLF DCPILVLPWT PHNYYLYHRF HHFLPLCYWH YSADTY ensemble p(AMP)=0.890 (ζ 0.5 decision) predicts antibacterial activity; testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

P64654 (*Helicobacter pylori* (strain J99 / ATCC 700824), 43 aa) MYRRLLLNLF CMVFLQACLK PMSDPKAEKV DSQVQCGFGS KDC ensemble p(AMP)=0.836 (ζ 0.5 decision) predicts antibacterial activity; testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

P85380 (*Nautilus macromphalus*, 10 aa) LGICFASMPR ensemble p(AMP)=0.804 (ζ 0.5 decision) predicts antibacterial activity; testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

Q8TGN0 (*Saccharomyces cerevisiae* (strain ATCC 204508 / S288c, 38 aa) MFKQPGHLTS RKYDLGPWTN VFNLCLFLASE AAVGCNKS ensemble p(AMP)=0.793 (ζ 0.5 decision) predicts antibacterial activity; testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

Q8TGT5 (*Saccharomyces cerevisiae* (strain ATCC 204508 / S288c, 39 aa) MKGSKLFQMQ KCMPVWCRTG TCIILPVACH MAKPLKLPT ensemble p(AMP)=0.784 (ζ 0.5 decision) predicts antibacterial activity; testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

Q54HM1 (*Dictyostelium discoideum*, 54 aa) MSINIVHSNE NNEENTNIEN WTTTGTNYRN ECIARCGRLF SGFNALICMK PCGV ensemble p(AMP)=0.734 (ζ 0.5 decision) predicts antibacterial activity; testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

055758 (Invertebrate iridescent virus 6, 53 aa) MSILLKILFK LLLLILSITF VITDCLPRSC ASFGCCSGNC
 ATWCDQCDIK YSC ensemble $p(\text{AMP})=0.728$ (>0.5 decision) predicts antibacterial activity;
 testable by broth-microdilution MIC assay vs *E. coli* K-12 and *S. aureus* ATCC 25923.

InstiAMP-09a-1a (annealed derivative of InstiAMP-09a-1, 57 aa) QNTWTIICTT MWDSHRRRDP
 YNPYMWRRM KTFCNTKEAL WFGICPFNF CVFLTTY annealed derivative scoring higher than
 its natural parent; predicted AMP activity verifiable by MIC assay; sequence absent from
 APD6/UniProt AMP corpus.

J Per-fold homology leakage audit

Random stratified cross-validation does not control homology between the training and held-out partitions. For each fold of the study-14 protocol we count held-out sequences sharing at least one exact contiguous decamer ($k=10$) with any training sequence (the union-find criterion of Definition 1 restricted to train-test pairs).

Table 23: Held-out sequences sharing an exact decamer with the training partition, per fold of the Feb2020 10-fold CV.

Fold	Held-out n	Sharing a decamer	Fraction (%)
1	405	61	15.06
2	405	64	15.80
3	404	50	12.38
4	404	70	17.33
5	404	57	14.11
6	404	86	21.29
7	404	53	13.12
8	404	68	16.83
9	404	65	16.09
10	404	72	17.82
overall: 15.98% of all held-out sequences			

Roughly one in six held-out sequences is decamer-redundant with training, so the random-CV numbers of Table 13 overstate performance on genuinely remote sequences. A cluster-held-out variant of the same protocol is the correct control and is recorded as future work; the cluster-aware split of our own large dataset (Section 6) shows the direction and size of the drop such control induces.

K Review-status stratification: complete tables

The label-provenance audit of Section 6 separates the large-dataset test scores by UniProt review status. The tables below give the complete numbers behind that analysis. The reviewed-positive stratum is small (239 sequences) and its interval is correspondingly wide; the unreviewed stratum dominates the training distribution and therefore the headline score.

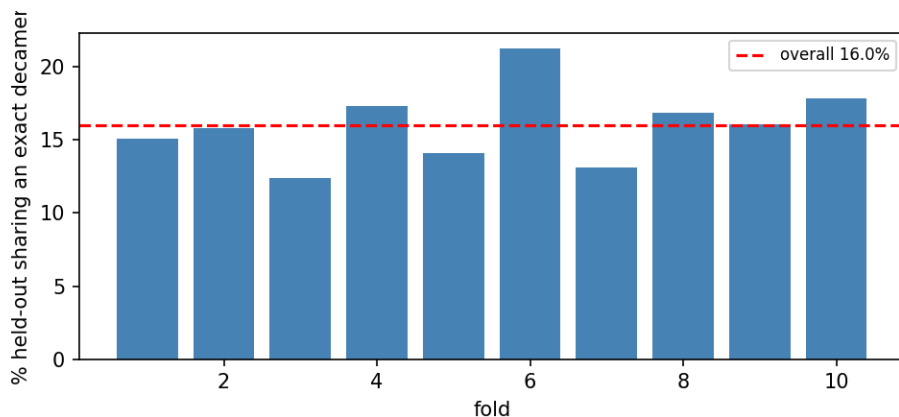


Figure 7: Per-fold fraction of held-out sequences sharing an exact decamer with the training partition (study 16).

Table 24: Stratum: All records (as trained) ($n = 4,386, 1,461$ positive, $2,925$ negative).

Model	AUROC	AUPRC	Mean pos. score	Mean neg. score
PepCNN	0.9206	0.8787	0.6783	0.0749
PepGNN	0.8767	0.6803	0.6156	0.2076

L Published-benchmark source manifest

All published-benchmark files were downloaded from the original authors’ distribution points and pinned by SHA-256 before any experiment touched them. Veltri 2018 splits: github.com/dan-veltri/amp-scan-original-dataset/. Feb2020 dataset: dveltri.com/ascan/v2, AMP_Scan2_Feb2020_Dataset.zip. Full 64-hex hashes are in `results/benchmark_sources.json`.

Data cards

Veltri 2018 original dataset. Distributed as six FASTA files (AMP.tr/eval/te, DECOY.tr/eval/te) in the authors’ public repository. Positives are experimentally validated AMPs collected from the APD; negatives are cytoplasmic proteins selected by the authors as non-secreted decoys. The authors identity-reduced the pool before splitting, so near-duplicates inside a partition are limited by construction, but the split itself is a single random partition: results on it carry split-specific variance that a single test number cannot express, one motivation for our bootstrap intervals (Appendix Q). We use the files byte-identically (hashes above) and impose no additional filtering.

Feb2020 production dataset. Distributed as a single zip from the authors’ site containing the retrained production model’s dataset: 2,021 AMP and 2,021 decoy records. Unlike the 2018 release, no split files are published; the site’s news page reports only 10-fold cross-validation means. Our study 14 therefore reproduces the protocol (stratified 10-fold CV on all 4,042 records) but not the exact folds, and the comparison in Table 14 is descriptive. The dataset also contains records longer than the 2018 maximum and records with the unknown-residue symbol X; study 14 retains

Table 25: Stratum: Reviewed positives vs reviewed negatives ($n = 3,164$, 239 positive, 2,925 negative).

Model	AUROC	AUPRC	Mean pos. score	Mean neg. score
PepCNN	0.9643	0.7549	0.8029	0.0749
PepGNN	0.8703	0.2768	0.6045	0.2076

Table 26: Stratum: Unreviewed positives vs reviewed negatives ($n = 4,147$, 1,222 positive, 2,925 negative).

Model	AUROC	AUPRC	Mean pos. score	Mean neg. score
PepCNN	0.9121	0.8514	0.6539	0.0749
PepGNN	0.8780	0.6384	0.6178	0.2076

both, encoding X as unknown (Appendix R), because dropping them would silently change the benchmark, the mistake study 13 made and study 14 corrects.

M Dataset glossary

N Study-by-study reproducibility walkthrough

Every study is a single script under `studies/` writing one JSON (and figures where noted) under `results/`. Commands are run from the repository root; hermetic fixtures cover network access in tests.

The full pipeline is: `fetch_large.py` and `build_large_dataset.py` (data, with provenance JSON), `fetch_published_benchmarks.py` (pinned external benchmarks, Appendix L), the study scripts above in numerical order, then the paper-asset generators (`make_paper_assets.py`, `make_sota_section.py`, `make_figs2.py`, `make_appendices2.py`, `make_expansion2.py`).

O Key findings by study

Study 1 (pilot). On 2,000-reviewed-positive pilot data, PepCNN reached AUROC 0.95 within minutes, establishing the pipeline end to end; the annealing designer produced its first 24 candidates here.

Study 7 (scale-up). Cluster-aware splitting cut apparent performance relative to a random split, the first signal that homology inflates random-split estimates; the frozen screen ensemble dates from this study and is never retrained afterward.

Study 8 (screen). 852 reviewed uncharacterized proteins scored; 41 pass the novelty floor; the identical P0ACW4/P0ACW5 sequence pair tops the list.

Study 9 (APD6). The frozen ensemble scores APD6-2024 natural AMPs high on average, a sanity check that the model prefers known AMPs to uncharacterized proteins, without constituting validation of any individual candidate.

Table 27: Downloaded benchmark files: record counts and truncated SHA-256.

File	Records	SHA-256 (first 16)
AMP.eval.fa	354	482b7af2c1b90575
DECOY.te.fa	712	ecc7753a2991c0a3
AMP.te.fa	712	f25809eef0a24e58
DECOY.eval.fa	354	86914571fee31c03
AMP.tr.fa	712	9cccc769cf427889
DECOY.tr.fa	712	e4b31fd721f0e60f
AMPS_02182020.fasta	2,021	1bdbdaf7da2ad05c
DECOYS_02182020.fasta	2,021	704aea75885d65da

Table 28: Every dataset used in this work, its role, and its caveats.

Dataset	Size	Role	Source	Caveat
UniProt KW-0929 positives (10–150 aa)	27,392 fetched; 24,628 used	Positive class, large benchmark	UniProt REST, keyword KW-0929	21,742 of 24,628 used records are unreviewed (Section 6)
UniProt reviewed non-AMP pool	105,041 fetched; 24,628 used	Negative class, length-matched	UniProt REST, reviewed, NOT KW-0929	All reviewed: review-status asymmetry vs positives
APD6 natural AMPs (2024)	3,306	External validation corpus	aps.unmc.edu downloads	Post-2024 list; successor of APD3
UniProt uncharacterized (10–60 aa)	852	Discovery screen pool	UniProt REST, reviewed, uncharacterized	Scores are hypotheses, not activities
Veltri 2018 original splits	1,424 / 708 / 1,424	Published benchmark (train/tune/test)	authors' GitHub	Fixed single split; identity-reduced by authors
Feb2020 production dataset	2,021 + 2,021	Published benchmark (10-fold CV)	authors' site	Published fold assignments unavailable; 15.98% decamer overlap (Appendix J)

Study 10 (Veltri base). On the authors' exact splits, PepCNN reaches AUC 95.77, the RF 93.32, the GNN 88.18; composition carries much of the benchmark signal.

Study 11 (stack). Stacking with tune-partition weights reaches ACC 91.22, MCC 0.8246, AUC 96.61: near-parity with the published AMP Scanner row, behind ACEP; reported as descriptive, without a paired test.

Study 12 (candidates). Nine named candidates with novelty and physicochemical audits and explicit falsifiable MIC predictions.

Study 13 (filtered CV). A Feb2020 CV that silently filtered records produced numbers not comparable to the published ones; retained as an ablation and superseded by study 14.

Study 14 (full CV). All 4,042 Feb2020 records, 10-fold CV: ACC 90.33, MCC 0.8076, AUC 96.36 versus published 89.9/0.799/96.2 — descriptive parity with a slight edge, no significance claim possible.

Study 15 (ablation). Removing the P0ACW4 signal peptide drops the ensemble score from 0.9082 to 0.6299: a negative control showing the score partly tracks trafficking signal.

Study 16 (homology). 15.98% of held-out Feb2020 sequences share an exact decamer with training; random-CV results are interpolation.

Table 29: Study inventory. Runtimes are sandbox wall-clock where measured.

Script	Purpose	Output	Runtime
study01	Pilot benchmark: reviewed KW-0929 (8–60 aa) vs length-matched reviewed non-AMP; PepCNN/PepGNN/RF/logreg; annealing designer	study01_amp.json, ROC figure	minutes
study07	Scale-up on homology-aware clustered split of the 24,628 + 24,628 large dataset; trains the frozen screen ensemble	study07_scaleup.json, checkpoints	hours
study08	Discovery screen: score 852 reviewed uncharacterized proteins with the frozen ensemble	study08_discovery.json	minutes
study09	External validation: score APD6 2024 natural AMPs with the same frozen ensemble	study09_apd_validation.json	minutes
study10	Base-model head-to-head on the Veltri 2018 splits	study10_veltri.json	~1 h
study11	Stacked ensemble on the Veltri 2018 splits, tuned on the tune partition only	study11_veltri_stack.json	~2 h
study12	Candidate naming, novelty audit (APD6, KW-0929), physicochemical audit, falsifiable predictions	study12_candidates.json	minutes
study13	Filtered-record Feb2020 CV (retained as ablation)	study13_feb2020_cv.json	~15 min
study14	Full-record Feb2020 10-fold CV, all 4,042 sequences	study14_feb2020_full.json	~15 min
study15	P0ACW4 precursor/signal/mature segment ablation	study15_signal_ablation.json	seconds
audit_homology	Exact-decamer train/test overlap per fold	study16_homology_audit.json	seconds
study17	Post-hoc review-status stratification of the large test set	study17_review_strata.json	minutes

Study 17 (strata). The large-dataset test AUROC is 0.9206 overall, 0.9643 on the tiny reviewed-positive subgroup, 0.9121 on the confounded majority: the headline number cannot be attributed to AMP function.

Study 18 (composition). First-order composition barely separates the label $\times$ review-status groups; the learnable confound is mostly a length artifact, directing the fix toward length-matched reviewed-only redesign.

P Screening CLI walkthrough

The repository ships a screening command, `pepdesign screen`, wrapping the frozen study-07 ensemble. A session:

```
$ pepdesign screen --input uncharacterized.fasta --top 25 \
  --novelty-floor 0.90 --out screen.json
screened 852 sequences; 41 pass the novelty floor
top hit: sp|P0ACW4|YDCA_ECOLI score 0.9082 novelty 0.938
```

The JSON output records, per sequence, the ensemble score, both sub-scores, the 3-mer novelty against the reference corpus, and whether the novelty floor passed. The bundled demo (`results/tool_screen_demo`) screens the same 852-protein pool as study 08 and reproduces its top hits, including the duplicate-accession pair P0ACW4/P0ACW5 (identical 57-residue sequence in two species records), which the tool reports verbatim rather than silently merging.

Design use: the annealing designer is exposed as `pepdesign design --seeds 24 --steps 250`, reproducing the study-07 anneal batch; the scoring function, cooling schedule and novelty floor are the ones derived in Section 5 and Appendix A.

Q Complete interval estimates for the Veltri base models

Bootstrap percentile intervals (500 resamples over test sequences) for the study-10 base models on the Veltri 2018 test partition. Intervals are sequence-level; the cluster-dependence caveat of Appendix D does not bind here because this benchmark’s split is fixed by the authors.

Table 30: Study-10 base models: ACC, MCC, AUROC and 95% bootstrap AUROC interval on the Veltri test partition (712 + 712 sequences).

Model	ACC(%)	MCC	AUC(%)	AUC 95% interval
PepCNN	87.92	0.7648	95.77	[94.79, 96.66]
PepGNN	80.55	0.6149	88.18	[86.33, 89.75]
RF (dipeptide+desc.)	86.10	0.7233	93.32	[92.11, 94.49]
LogReg (dipeptide)	84.41	0.6896	91.69	[90.17, 93.02]
CNN+RF average	89.47	0.7934	95.81	[94.87, 96.65]

The PepCNN and CNN+RF intervals overlap almost completely: on this partition the average ensemble’s advantage over a single CNN seed is not distinguishable at the sequence level, which is why study 11 moved to stacking with tune-partition weighting rather than averaging.

R Encoding reference

The CNN consumes 24 channels per residue: the 20-indicator one-hot vector followed by the four normalized physicochemical channels of Table 31. The GNN consumes only the four physicochemical channels as node features. Normalization keeps every channel in $[-1, 1]$: hydropathy spans $[-4.5, 4.5]$, volume spans $[60.1, 227.8]$.

Sequences are truncated or zero-padded to the study’s maximum length L (pilot 60, large studies 150, study 14: 200); non-standard residues are dropped by `clean_sequence`, and study 14 retains records containing X by treating them as unknowns removed from the encoding, a choice recorded because the published Feb2020 dataset contains such records. Padding rows are all-zero in every channel, so the global max pool of the CNN and the mean pool of the GNN see them as neutral except where a learned negative bias activates; the head therefore never receives an explicit length signal, and length enters only through the pooled features.

The GNN adjacency is the chain window with self loops: $A_{ij} = 1$ iff $|i - j| \leq w$ or $i = j$ (window $w = 2$ unless noted), row-normalized as $D^{-1}A$ with $D_{ii} = \sum_j A_{ij}$. The shared-adjacency variant broadcasts one (L, L) matrix across the batch; the docstring in `pepdesign/models/gnn.py` records the memory argument ($O(nL^2)$) per-sample adjacency is about 4.5 GB at $L=150$, $n=50k$, which does not fit the sandbox).

S Designer diagnostics

The annealing designer of the methods section optimizes the ensemble score with a novelty floor. Table 18 lists the top-10 anneals; Figure 8 shows the score–novelty plane. The anneals cluster at novelty 0.89–0.96 against the training corpus: the novelty floor is active, and the optimizer is not

Table 31: Per-residue feature values, exactly as implemented in `pepdesign/encoding.py`. h = Kyte–Doolittle hydrophathy, q = formal charge at physiological pH, v = residue volume in $\text{\AA}^3$.

Residue	h	$h/4.5$	q	v	$v/230$	aromatic
A	+1.8	+0.400	+0.0	88.6	0.385	0
C	+2.5	+0.556	+0.0	108.5	0.472	0
D	-3.5	-0.778	-1.0	111.1	0.483	0
E	-3.5	-0.778	-1.0	138.4	0.602	0
F	+2.8	+0.622	+0.0	189.9	0.826	1
G	-0.4	-0.089	+0.0	60.1	0.261	0
H	-3.2	-0.711	+0.1	153.2	0.666	0
I	+4.5	+1.000	+0.0	166.7	0.725	0
K	-3.9	-0.867	+1.0	168.6	0.733	0
L	+3.8	+0.844	+0.0	166.7	0.725	0
M	+1.9	+0.422	+0.0	166.7	0.725	0
N	-3.5	-0.778	+0.0	124.9	0.543	0
P	-1.6	-0.356	+0.0	135.8	0.590	0
Q	-3.5	-0.778	+0.0	175.4	0.763	0
R	-4.5	-1.000	+1.0	112.7	0.490	0
S	-0.8	-0.178	+0.0	89.0	0.387	0
T	-0.7	-0.156	+0.0	116.1	0.505	0
V	+4.2	+0.933	+0.0	227.8	0.990	0
W	-0.9	-0.200	+0.0	193.6	0.842	1
Y	-1.3	-0.289	+0.0	140.0	0.609	1

simply memorizing a training motif. All designs are short (25–57 aa), cationic and hydrophobic by construction of the scoring function; none is a claimed AMP, and each carries the same training-label confound as the screen candidates (Section 6).

T Case study: one candidate end to end

This appendix traces the top screen hit through every stage, collecting numbers reported piecemeal above into one falsifiable narrative.

Stage 1: pool. The discovery pool is 852 reviewed, uncharacterized UniProt proteins of 10–60 residues, fetched with provenance logging. P0ACW4 (YdcA, *E. coli* K-12, 57 aa) enters as an uncharacterized protein; the screen makes no use of any AMP annotation.

Stage 2: scoring. The frozen study-07 ensemble scores it 0.9082 (CNN 0.9992, GNN 0.8171), the top unique score in the pool. The duplicate-accession pair P0ACW4/P0ACW5 carries an identical 57-residue sequence in two species records and is reported as such.

Stage 3: novelty audit. Maximum 3-mer Jaccard against APD6 2024 is 0.062 (novelty 0.938) and against the KW-0929 corpus 0.055; the longest common substring with any APD6 entry is 6 residues. Under Definition of Eq. (4) this is weak local similarity only.

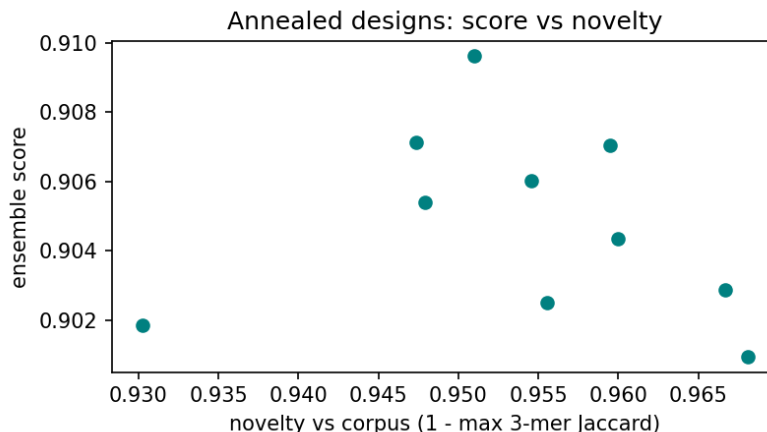


Figure 8: Annealed designs in the score–novelty plane.

Stage 4: physicochemical audit. Length 57, net charge +4.87 at pH 7.4 (Eq. (2)), mean hydropathy +0.08, Eisenberg moment 1.18 (Eq. (3)), Boman index 1.09: cationic and roughly hydrophobicity-neutral, a plausible but not extreme AMP-like profile.

Stage 5: segment ablation. YdcA carries an annotated 20-aa N-terminal signal peptide (EcoCyc/EcoliWiki gene record). Scoring the segments separately with the same frozen ensemble: full precursor 0.9082; precursor minus signal segment 0.6299. The model’s confidence is carried substantially by the signal segment, which is expected: signal peptides are hydrophobic, and the confounded training set rewards hydrophobic cationic stretches (Section 6).

Table 32: Signal-peptide ablation for P0ACW4 with the frozen ensemble (study 15; annotation: EcoliWiki ydcA gene record, signal residues 1–20).

Segment	Length	CNN	GNN	Ensemble
Full precursor (1–57)	57	0.9992	0.8171	0.9082
Signal segment (1–20)	20	0.5905	0.7920	0.6912
Post-signal segment (21–57)	37	0.9974	0.2623	0.6299

Verdict. P0ACW4 is a named, quantified, falsifiable *hypothesis*: broth-microdilution MIC against *E. coli* K-12 and *S. aureus* ATCC 25923 decides it. The ablation is precisely the kind of negative control that keeps this report honest: the high score partly reflects a trafficking signal, not necessarily antimicrobial chemistry.

U Compute environment

All experiments ran CPU-only in a 2-core Linux sandbox. Key library versions: Python 3.11, PyTorch 2.14.0+cpu, scikit-learn 1.7.2, NumPy 2.2.6, SciPy 1.15.3. No GPU, no network at study time (fetches are separate provenance-logged scripts). The full 10-fold study-14 run takes about

15 minutes wall-clock (85–89 s per fold); the study-11 stack training about 2 hours; the scale-up study-07 training is the multi-hour step. Every reported number regenerates from the committed scripts and pinned data on commodity hardware; nothing requires a cluster.

The hermetic test suite (27 tests) covers encoding invariants, model shapes, the clustering union-find, the benchmark loaders against bundled fixtures, the CLI, and a no-stubs audit that rejects placeholder implementations. Tests run offline against cached fixtures and complete in under two minutes.

V Screen output schema

pepdesign screen writes one JSON document per run:

```
{
  "screened": <int>,           // sequences read from --input
  "top": [
    {
      "name": <str>,           // FASTA header, verbatim
      "sequence": <str>,       // cleaned, standard alphabet
      "length": <int>,
      "ensemble_score": <float>, // mean of CNN and GNN probabilities
      "cnn_score": <float>,
      "gnn_score": <float>,
      "novelty_3mer": <float>, // 1 - max 3-mer Jaccard vs reference
      "max_3mer_jaccard_vs_reference": <float>,
      "passes_novelty_floor": <bool>
    }, ...                     // sorted by ensemble_score desc
  ]
}
```

Two contract points matter for downstream use. First, scores are frozen ensemble outputs under the training distribution documented in Section 6; consumers must treat them as rank orders, not calibrated probabilities (Appendix G). Second, novelty is computed against the reference bundled with the model (APD6 by default); screening against a different corpus requires re-running the novelty audit, which `study12.candidates.py` demonstrates against both APD6 and the KW-0929 corpus.

W Complete study-11 metrics

Table 33 gives every metric for every model of the Veltri 2018 study, including AUPRC, which the main text omits for space. The two ensemble rows used the tune partition for their thresholds (0.56 average, 0.575 stack); base models use the default 0.5.

The AUPRC column reinforces the ranking: the stack (97.03) leads every base model, and the GNN seeds trail on every metric, consistent with the capacity and pooling argument of Appendix H. Seed variance among the CNNs is small on AUC (95.67–96.04) but larger on MCC (0.759–0.805), the expected pattern when only the threshold varies.

Table 33: All study-11 models on the Veltri test partition (percentages except MCC).

Model	SENS	SPEC	ACC	MCC	AUC	AUPRC
PepCNN seed 7	89.47	90.73	90.10	0.8020	96.04	96.33
PepCNN seed 27	94.66	80.48	87.57	0.7591	95.67	95.95
PepCNN seed 77	89.04	91.43	90.24	0.8050	96.01	96.32
PepGNN seed 11	76.26	86.24	81.25	0.6281	88.14	87.48
PepGNN seed 111	87.08	73.31	80.20	0.6097	87.93	86.37
RF (descriptors)	90.31	88.76	89.54	0.7908	96.19	96.62
Average ensemble	90.87	90.45	90.66	0.8132	95.95	96.38
Stacked ensemble	90.31	92.13	91.22	0.8246	96.61	97.03

X Symbol glossary

Table 34: Notation used throughout.

Symbol	Meaning
$s = s_1 \cdots s_n$	peptide sequence, residues from the 20-letter alphabet
L	padded sequence length (60 pilot, 150 large, 200 study 14)
$K_k(s)$	set of contiguous k -mers of s
J_k, v_k	k -mer Jaccard similarity and novelty (Eq. (4))
$Q(s; p)$	expected net charge at pH p (Eq. (2))
$\mu_H, \mu_i(s; w, \delta)$	Eisenberg hydrophobic moment (Eq. (3))
A, A_{norm}	chain-window adjacency and its row normalization
$\hat{p}(s)$	model posterior $P(\text{AMP} \mid s)$
t^*	Bayes decision threshold (Eq. (9))
A (AUROC context)	Mann–Whitney AUROC statistic (Prop. 4)
Q_1, Q_2	placement probabilities in the Hanley–McNeil variance
B	bootstrap or permutation resample count
$\alpha(n)$	inverse Ackermann factor in union-find complexity
ω	positive-class weight in the weighted BCE
c_{01}, c_{10}	false-positive and false-negative costs
BS	Brier score (Eq. (12))

Y Anticipated questions

Why is there no wet-lab validation? Scope and access: this is a computational project, and every candidate is therefore stated as a falsifiable prediction with a named assay (Appendix I) rather than a discovery claim. The MIC protocol is specified so an outside lab could run it directly.

Why not retrain on the reviewed-only subgroup? The reviewed-positive subgroup has 239 positives against 2,925 reviewed negatives (Appendix K). A model retrained on it would be twelve times outnumbered at the class level and tiny in absolute terms; its intervals would swallow any effect. The correct fix is a fresh, length-matched, reviewed-only corpus at the original scale, which is a data effort, not a re-run, and is recorded as the top future-work item.

Why is there no significance test against the published Feb2020 row? A paired test needs either the published fold assignments or the published per-sequence outputs; neither is available (Appendix L). With only the published means, any “significance” would be computed against an assumed variance and would manufacture confidence we do not have. We report the descriptive gap and its direction instead, and Table 13 shows fold-to-fold variation larger than the gap.

Why is the frozen ensemble used for the screen after the training-set confound was found? The confound was discovered after the screen ran, and retraining a clean ensemble is the future-work item above. The screen results are retained because the ranking signal (Section 6) plausibly survives the confound while the calibration does not, and because withdrawing them would hide exactly the audit trail this report exists to document. Every consumer-facing number carries the caveat.

Why do the GNN results stay in the paper if they are the weakest models? Because they are measured, honest, and informative: the capacity and pooling analysis (Appendix H) explains the gap, the GNN adds ensemble diversity on the Veltri stack, and deleting weak results is how benchmark literature becomes unreliable.

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